

Über die logische Notwendigkeit Gottes und die Unmöglichkeit des Teufels in Gödels modalem ontologischem Argument

Erkenntnisse aus Experimenten in der Computationalen Metaphysik

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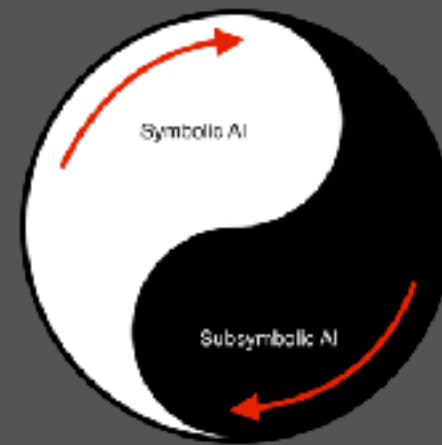


Berlin-Brandenburgische Akademie der Wissenschaften, Berlin, 30. Juli 2026

Einführende Bemerkungen

Computationale Metaphysik

Formale Methoden zur Analyse philosophischer Argumente



Analyse philosophischer Argumente mithilfe formaler Methoden

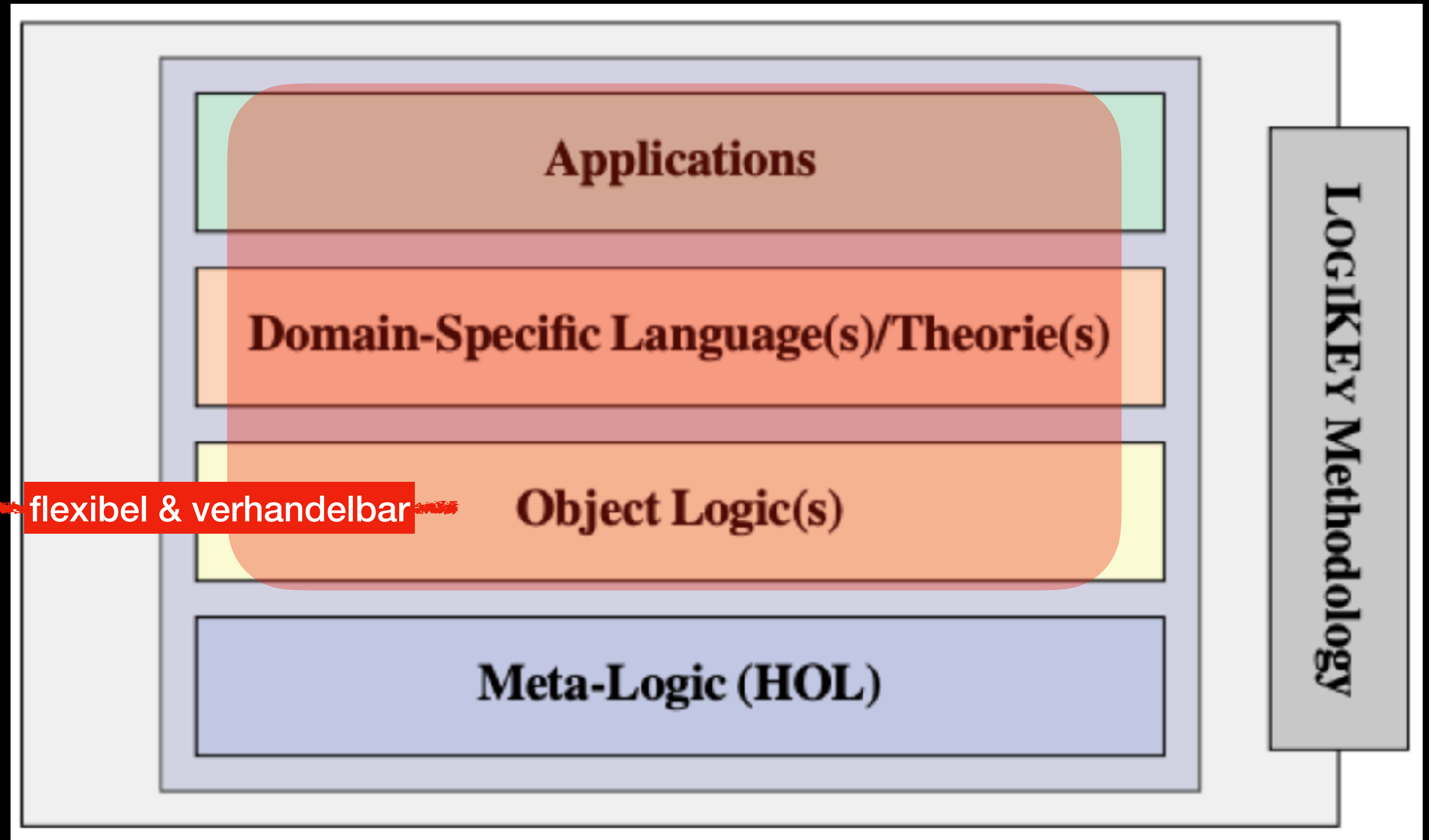
Logikpluralistische Wissensrepräsentation und Schließen

LogiKEy

Unterschiedliche Modallogiken:
K, T, KB, S4, S5, etc

Unterschiedliche Quantoren
Extensional oder Intensional
usw.

flexibel & verhandelbar



Anfänge: 2007-2008 (mit L. Paulson)

Begriff LogiKEy: 2017-2019 (mit L. van der Torre & X. Parent)

'Exakte' Methoden sind möglich ...



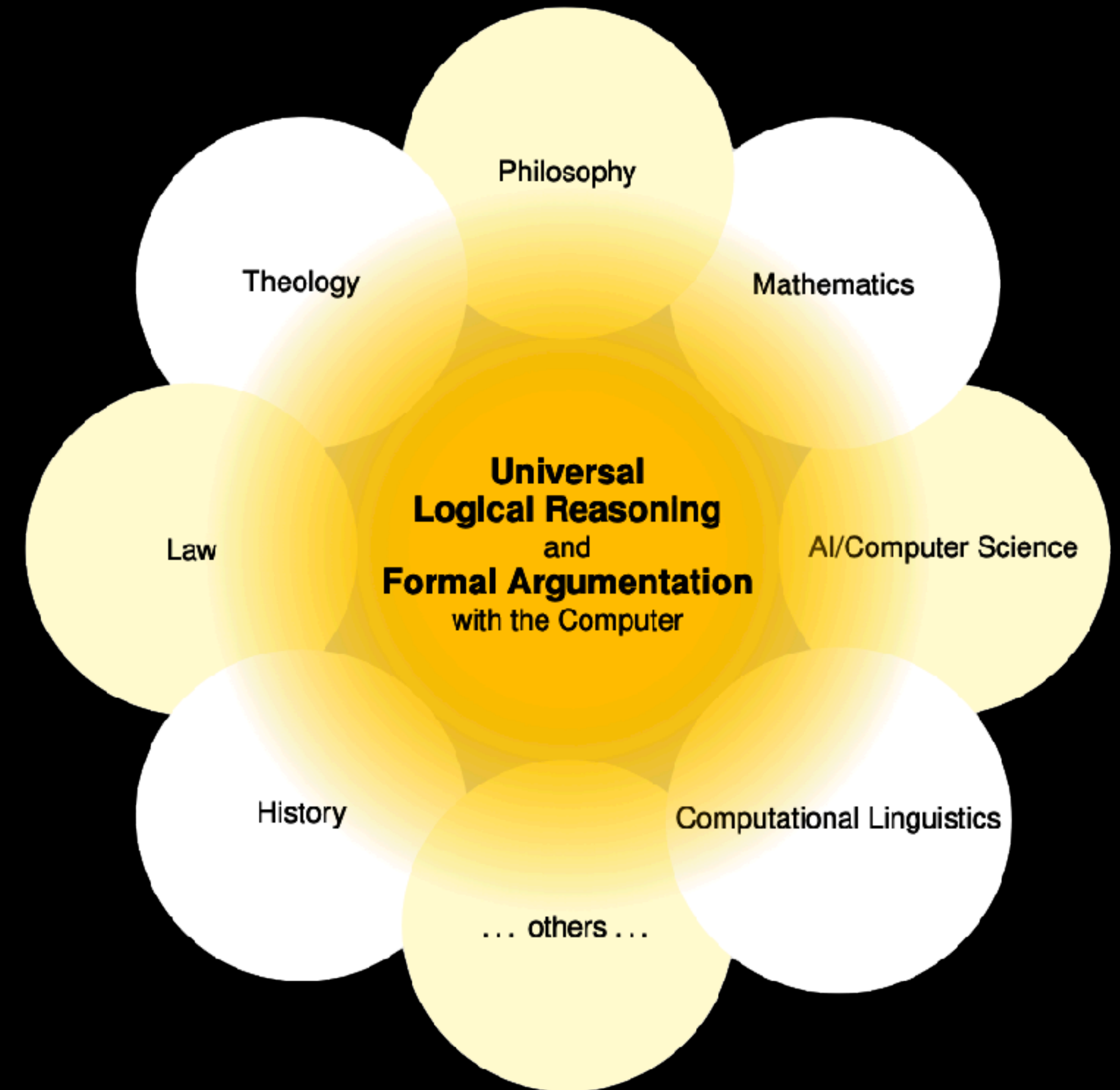
Es gibt systematische Methoden zur Lösung aller Probleme (auch Kunst etc.).

Es gibt eine wissenschaftliche (exakte) Philosophie {und Theologie}, welche die Begriffe der höchsten Abstraktheit behandeln.

Kurt Gödel: Maximilian V



Kurt Gödel (1906-78)



... über die Grenzen der Mathematik hinaus!

Anwendung von LogiKEy auf Gödels modales ontologisches Argument

Arbeiten, die ich hier nicht anspreche:

- Analyse von Scotts Variante
- Analyse von Fittings Variante (**Extensionen**)
- Analyse von Andersons Variante
- Studien zu verwandten Argumenten
- Studien zu (kontrovers) simplifizierten Varianten

Types, Tableaus and Gödel's God in Isabelle/HOL 2017
David Fuenmayor, Christoph Benzmüller · *Archive of Formal Proofs (data publication)*, 1–34

Automating Emendations of the Ontological Argument in Intensional Higher-Order Modal Logic 2017
David Fuenmayor, Christoph Benzmüller · *KI 2017: Advances in Artificial Intelligence, 40th Annual German Conference on AI, Dortmund, Germany, September 25-29, 2017, Proceedings* pp. 114-127

Computer-supported Analysis of Positive Properties, Ultrafilters and Modal Collapse in Variants of Gödel's Ontological Argument 2020
Christoph Benzmüller, David Fuenmayor · *Bulletin of the Section of Logic* 49(2), 127-148

A Case Study On Computational Hermeneutics: E. J. Lowe's Modal Ontological Argument 2020
David Fuenmayor, Christoph Benzmüller · *Beyond Faith and Rationality: Essays on Logic, Religion and Philosophy*

A Simplified Variant of Gödel's Ontological Argument 2023
Christoph Benzmüller · *Beyond Babel: Religions and Linguistic Pluralism*

...

Wichtig: In diesem Vortrag betrachten wir **modale Logik(en) als Erweiterung(en) klassischer Logik, Typentheorie (Church), volle Komprehension, Intensionen von Eigenschaften**

Rest dieser Präsentation basiert auf ...

Automating Gödel's Ontological Proof of God's Existence with Higher-order Automated Theorem Provers
Christoph Benz Müller, Bruno Woltzenlogel Paleo · *ECAI 2014* pp. 93 – 98 2014

The Inconsistency in Gödel's Ontological Argument: A Success Story for AI in Metaphysics
Christoph Benz Müller, Bruno Woltzenlogel Paleo · *IJCAI 2016* pp. 936-942 2016

arXiv

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Computer Science > Logic in Computer Science

[Submitted on 26 May 2026]

Many Logics, One Methodology: A Plea for Logical Pluralism in Formalised Reasoning (preprint)

Christoph Benz Müller, Daniel Kirchner, Luca Pasetto

This position statement looks back on two decades of work on shallow embeddings of non-classical logics in classical higher-order logic (HOL), a line of research that expanded into a range of logic embeddings in HOL and inspired the LogiKey logic-pluralistic knowledge representation and reasoning methodology. This paper advances the case for logical pluralism at object-logic level within a unifying meta-logical framework such as LogiKey, grounding the argument in computational metaphysics. More broadly, it advocates principled support for logical pluralism in modern proof assistants, and cautions against logical imperialism -- the rigid adoption of a single foundational logic for large-scale theory developments -- which impedes the interdisciplinary reuse that LogiKey is designed to enable.

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Notes on Gödel's and Scott's variants of the ontological argument

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Volume 208, pages 569–611 (2025) [Cite this article](#)

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Christoph Benz Müller  & Dana Scott

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Abstract

Notes on Kurt Gödel's modal ontological argument and Dana Scott's variant of it are presented. These remarks, supported by experimental studies with a proof assistant system for classical higher-order logic, implicitly answer some questions the authors have received over the last decade(s). In addition, some new insights resulting from the conducted experiments are reported.

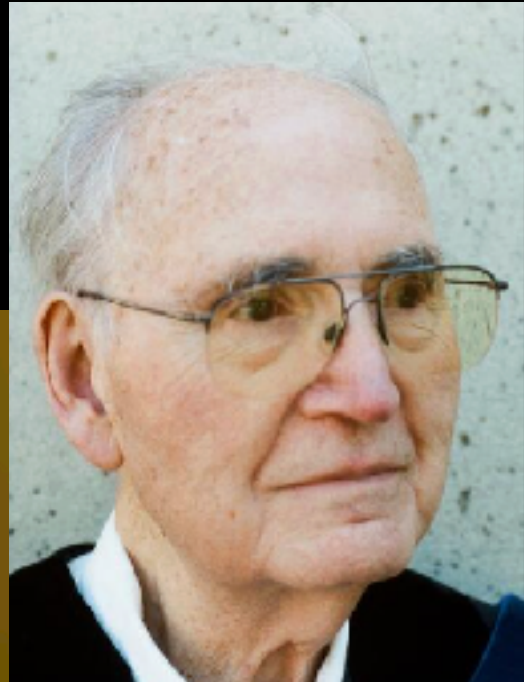
Monatshefte für Mathematik

Volume 171 - Numbers 3-4 - 2013



Monatshefte für Mathematik

[Aims and scope](#) >
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gem. mit: Dana Scott

... und einigen weiteren, neuen Einsichten

Ontologischer Beweis

Feb 10, 1970

$P(\varphi)$ φ is positive ($\varphi \in P$)

At 1 $P(\varphi) \cdot P(\psi) \supset P(\varphi \cdot \psi)$ At 2 $P(\varphi) \vee P(\sim \varphi)$

P1 $G(x) \equiv (\varphi) [P(\varphi) \supset \varphi(x)]$ (God)

P2 $\varphi \text{ Ess } x \equiv (\psi) [\psi(x) \supset N(y) [\varphi(y) \supset \psi(y)]]$ (Essence of x)

$p \supset_N q = N(p \supset q)$ Necessity

At 2 $P(\varphi) \supset NP(\varphi)$
 $\sim P(\varphi) \supset N \sim P(\varphi)$ } because it follows from the nature of the property

Th. $G(x) \supset G \text{ Ess } x$

Df. $E(x) \equiv (\varphi) [\varphi \text{ Ess } x \supset N \exists x \varphi(x)]$ necessary Existence

Ax 3 $P(E)$

Th. $G(x) \supset N(\exists y) G(y)$

hence $(\exists x) G(x) \supset N(\exists y) G(y)$

" $M(\exists x) G(x) \supset MN(\exists y) G(y)$
 " $\supset N(\exists y) G(y)$ M = possibility

any two essences of x are nec. equivalent

exclusive or and for any number of summands **wichtig!**

höherstufig

modal

$M(\exists x) G(x)$ means ^{the system of} all pos. prop. is compatible
 This is true because of:

At 4: $P(\varphi) \cdot \varphi \supset_N \psi \supset P(\psi)$ which impl

~~hence~~ $\begin{cases} x=x & \text{is positive} \\ x \neq x & \text{is negative} \end{cases}$

But if a system S of pos. prop. were incomp. It would mean that the sum prop. S (which is positive) would be $x \neq x$

Positive means positive in the moral aesthetic sense (independently of the accidental structure of the world). Only then the ax. true. It also means "attribution" as opposed to "privation" (or containing privation). This interprets simpler proof

If φ pos. then $(x) N \sim \varphi(x)$ Otherwise $\varphi(x) \supset_N x \neq x$

hence $x \neq x$ positive so $x=x$ neg. contradictory At 4 of the exist. of pos. prop.

x i.e. the normal form in terms of elem. prop. contains a member without negation.

Überblick

- HOML in HOL
- ~~Scotts Variante (mit PG, wie wir es bereits kennen)~~
- Gödels Scriptum-Variante (1970)
 - a. **Gödel1**: Scriptum-Begriff der Essenz — Widerspruch
 - b. **Gödel2**: modifizierter Begriff von Essenz — $\Box \exists G$
 - c. **Gödel3**: modifizierter Begriff von Eigenschaftsinklusion — $\Box \exists G$
 - d. **Kein (perfekter) Teufel (oder Teufel = Gott)** — $\Box \exists G$
- Gödels Scriptum-Variante (1970) und mathematischer Realismus
— Gott ist (überabzählbar/maximal) unendlich
- Persönliches Fazit

neue formale Studien

neue Variante

LogiKEY@work

HOML in HOL

HOML in HOL



Unwichtige Isabelle/HOL Parametereinstellungen

```
1 theory HOMLinHOL imports Main
2 begin
3 —<Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4 nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5 declare[[syntax_ambiguity_warning=false]]
```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKEy methodology

HOML in HOL



Typ **i** für mögliche Welten

Typ **e** für Individuen

Typ $\sigma := (i \Rightarrow \text{bool})$ für modale Formeln

Typ $\tau := (e \Rightarrow \sigma)$ für modale Prädikate

Konstante **r** für die Erreichbarkeit zwischen Welten

r je nach Logik **refl./trans./symm.** (K, T, KB, S4, S5)

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5 declare[[syntax_ambiguity_warning=false]]
6 —<Type i is associated with possible worlds and type e with entities:>
7 typedecl i —<Possible worlds>
8 typedecl e —<Individuals/entities>
9 type_synonym  $\sigma$  = "i  $\Rightarrow$  bool" —<World-lifted propositions>
10 type_synonym  $\tau$  = "e  $\Rightarrow$   $\sigma$ " —<Modal properties>
11 consts R::"i  $\Rightarrow$  i  $\Rightarrow$  bool" ("_r_") —<Accessibility relation between worlds>
12 axiomatization where
13 Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
```

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HOML in HOL



Modallogische Konnektive $\perp$ $\top$ $\neg$ $\wedge$ $\vee$ $\supset$ $\leftrightarrow$

```

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2 begin
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13    Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
14  <Logical connectives (operating on truth-sets):>
15  abbreviation Mbot:: $\sigma$  ("⊥") where " $\perp \equiv \lambda w. \text{False}$ "
16  abbreviation Mtop:: $\sigma$  ("⊤") where " $\top \equiv \lambda w. \text{True}$ "
17  abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬_") [52]53) where " $\neg \varphi \equiv \lambda w. \neg(\varphi w)$ "
18  abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl " $\wedge$ " 50) where " $\varphi \wedge \psi \equiv \lambda w. \varphi w \wedge \psi w$ "
19  abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl " $\vee$ " 49) where " $\varphi \vee \psi \equiv \lambda w. \varphi w \vee \psi w$ "
20  abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\supset$ " 48) where " $\varphi \supset \psi \equiv \lambda w. \varphi w \longrightarrow \psi w$ "
21  abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl " $\leftrightarrow$ " 47) where " $\varphi \leftrightarrow \psi \equiv \lambda w. \varphi w \longleftrightarrow \psi w$ "

```

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HOML in HOL



Box und Diamond: $\Box$ $\Diamond$ (über Erreichbarkeit r)

```

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2 begin
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23 abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇_") [54]55) where " $\Diamond \varphi \equiv \lambda w. \exists v. w r v \wedge \varphi v$ "

```

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HOML in HOL



Primitive Gleichheit (rigide, polymorph)

```

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2 begin
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HOML in HOL



Komplement & Konjunktion modaler Eigenschaften,
und Exklusiv-Oder

```

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26  abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬~_") where "¬~ $\Phi \equiv \lambda x. \lambda w. \neg \Phi x w$ "
27  abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where " $\Phi . \Psi \equiv \lambda x. \lambda w. \Phi x w \wedge \Psi x w$ "
28  abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨^e" 49) where " $\varphi \vee^e \psi \equiv (\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ "

```

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HOML in HOL



Quantoren (konstante Domänen, polymorph)

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28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where " $\varphi \vee^e \psi \equiv (\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ "
29 —<Possibilist quantifiers (polymorphic)>
30 abbreviation Mallposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀") where " $\forall \Phi \equiv \lambda w. \forall x. \Phi x w$ "
31 abbreviation Mallpossb (binder "∀" [8]9) where " $\forall x. \varphi(x) \equiv \forall \varphi$ "
32 abbreviation Mexiposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃") where " $\exists \Phi \equiv \lambda w. \exists x. \Phi x w$ "
33 abbreviation Mexiposb (binder "∃" [8]9) where " $\exists x. \varphi(x) \equiv \exists \varphi$ "

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKEy methodology

HOML in HOL



Quantoren (variable Domänen, für Individuen)
($x@w$ modelliert die Existenz von x in Welt w)

```

1 theory HOMLinHOL imports Main
2 begin
3 —<Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4 nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5 declare[[syntax_ambiguity_warning=false]]
6 —<Type i is associated with possible worlds and type e with entities:>
7 typedecl i —<Possible worlds>
8 typedecl e —<Individuals/entities>
9 type_synonym  $\sigma$  = "i $\Rightarrow$ bool" —<World-lifted propositions>
10 type_synonym  $\tau$  = "e $\Rightarrow$  $\sigma$ " —<Modal properties>
11 consts R::"i $\Rightarrow$ i $\Rightarrow$ bool" ("_r_") —<Accessibility relation between worlds>
12 axiomatization where
13   Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
14 —<Logical connectives (operating on truth-sets):>
15 abbreviation Mbot:: $\sigma$  ("⊥") where " $\bot \equiv \lambda w. \text{False}$ "
16 abbreviation Mtop:: $\sigma$  ("⊤") where " $\top \equiv \lambda w. \text{True}$ "
17 abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬") [52]53) where " $\neg \varphi \equiv \lambda w. \neg(\varphi w)$ "
18 abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∧" 50) where " $\varphi \wedge \psi \equiv \lambda w. \varphi w \wedge \psi w$ "
19 abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨" 49) where " $\varphi \vee \psi \equiv \lambda w. \varphi w \vee \psi w$ "
20 abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃" 48) where " $\varphi \supset \psi \equiv \lambda w. \varphi w \longrightarrow \psi w$ "
21 abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "↔" 47) where " $\varphi \leftrightarrow \psi \equiv \lambda w. \varphi w \longleftrightarrow \psi w$ "
22 abbreviation Mbox::" $\sigma \Rightarrow \sigma$ " ("□") [54]55) where " $\Box \varphi \equiv \lambda w. \forall v. w r v \longrightarrow \varphi v$ "
23 abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇") [54]55) where " $\Diamond \varphi \equiv \lambda w. \exists v. w r v \wedge \varphi v$ "
24 abbreviation Mprimeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("=_") where " $x = y \equiv \lambda w. x = y$ "
25 abbreviation Mprimneg::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≠") where " $x \neq y \equiv \lambda w. x \neq y$ "
26 abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬~") where " $\sim \Phi \equiv \lambda x. \lambda w. \neg \Phi x w$ "
27 abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where " $\Phi . \Psi \equiv \lambda x. \lambda w. \Phi x w \wedge \Psi x w$ "
28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where " $\varphi \vee^e \psi \equiv (\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ "
29 —<Possibilist quantifiers (polymorphic)>
30 abbreviation Mallposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀") where " $\forall \Phi \equiv \lambda w. \forall x. \Phi x w$ "
31 abbreviation Mallpossb (binder "∀" [8]9) where " $\forall x. \varphi(x) \equiv \forall \varphi$ "
32 abbreviation Mexiposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃") where " $\exists \Phi \equiv \lambda w. \exists x. \Phi x w$ "
33 abbreviation Mexipossb (binder "∃" [8]9) where " $\exists x. \varphi(x) \equiv \exists \varphi$ "
34 —<Actualist quantifiers (for individuals/entities)>
35 consts existsAt::" $e \Rightarrow \sigma$ " ("_@_")
36 abbreviation Mallact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀E") where " $\forall^E \Phi \equiv \lambda w. \forall x. x@w \longrightarrow \Phi x w$ "
37 abbreviation Mallactb (binder "∀E" [8]9) where " $\forall^E x. \varphi(x) \equiv \forall^E \varphi$ "
38 abbreviation Mexiact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃E") where " $\exists^E \Phi \equiv \lambda w. \exists x. x@w \wedge \Phi x w$ "
39 abbreviation Mexiactb (binder "∃E" [8]9) where " $\exists^E x. \varphi(x) \equiv \exists^E \varphi$ "

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKey methodology

HOML in HOL



Leibniz-Gleichheit (polymorph)

```

1 theory HOMLinHOL imports Main
2 begin
3 —<Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4 nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5 declare[[syntax_ambiguity_warning=false]]
6 —<Type i is associated with possible worlds and type e with entities:>
7 typedecl i —<Possible worlds>
8 typedecl e —<Individuals/entities>
9 type_synonym  $\sigma$  = "i $\Rightarrow$ bool" —<World-lifted propositions>
10 type_synonym  $\tau$  = "e $\Rightarrow$  $\sigma$ " —<Modal properties>
11 consts R::"i $\Rightarrow$ i $\Rightarrow$ bool" ("_r_") —<Accessibility relation between worlds>
12 axiomatization where
13   Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
14 —<Logical connectives (operating on truth-sets):>
15 abbreviation Mbot:: $\sigma$  ("⊥") where " $\bot \equiv \lambda w. \text{False}$ "
16 abbreviation Mtop:: $\sigma$  ("⊤") where " $\top \equiv \lambda w. \text{True}$ "
17 abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬_") [52]53) where " $\neg \varphi \equiv \lambda w. \neg(\varphi w)$ "
18 abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∧" 50) where " $\varphi \wedge \psi \equiv \lambda w. \varphi w \wedge \psi w$ "
19 abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨" 49) where " $\varphi \vee \psi \equiv \lambda w. \varphi w \vee \psi w$ "
20 abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃" 48) where " $\varphi \supset \psi \equiv \lambda w. \varphi w \longrightarrow \psi w$ "
21 abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "↔" 47) where " $\varphi \leftrightarrow \psi \equiv \lambda w. \varphi w \longleftrightarrow \psi w$ "
22 abbreviation Mbox::" $\sigma \Rightarrow \sigma$ " ("□_" [54]55) where " $\Box \varphi \equiv \lambda w. \forall v. w r v \longrightarrow \varphi v$ "
23 abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇_" [54]55) where " $\Diamond \varphi \equiv \lambda w. \exists v. w r v \wedge \varphi v$ "
24 abbreviation Mprimeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("=_") where " $x = y \equiv \lambda w. x = y$ "
25 abbreviation Mprimneg::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≠_") where " $x \neq y \equiv \lambda w. x \neq y$ "
26 abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬~") where " $\sim \Phi \equiv \lambda x. \lambda w. \neg \Phi x w$ "
27 abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where " $\Phi . \Psi \equiv \lambda x. \lambda w. \Phi x w \wedge \Psi x w$ "
28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where " $\varphi \vee^e \psi \equiv (\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ "
29 —<Possibilist quantifiers (polymorphic)>
30 abbreviation Mallposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀") where " $\forall \Phi \equiv \lambda w. \forall x. \Phi x w$ "
31 abbreviation Mallpossb (binder "∀" [8]9) where " $\forall x. \varphi(x) \equiv \forall \varphi$ "
32 abbreviation Mexiposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃") where " $\exists \Phi \equiv \lambda w. \exists x. \Phi x w$ "
33 abbreviation Mexipossb (binder "∃" [8]9) where " $\exists x. \varphi(x) \equiv \exists \varphi$ "
34 —<Actualist quantifiers (for individuals/entities)>
35 consts existsAt::" $e \Rightarrow \sigma$ " ("@_")
36 abbreviation Mallact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀E") where " $\forall^E \Phi \equiv \lambda w. \forall x. x @ w \longrightarrow \Phi x w$ "
37 abbreviation Mallactb (binder "∀E" [8]9) where " $\forall^E x. \varphi(x) \equiv \forall^E \varphi$ "
38 abbreviation Mexiact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃E") where " $\exists^E \Phi \equiv \lambda w. \exists x. x @ w \wedge \Phi x w$ "
39 abbreviation Mexiactb (binder "∃E" [8]9) where " $\exists^E x. \varphi(x) \equiv \exists^E \varphi$ "
40 —<Leibniz equality (polymorphic):>
41 abbreviation Mleibeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≡") where " $x \equiv y \equiv \forall P. P x \supset P y$ "

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKEy methodology

HOML in HOL



Gültigkeitsbegriff für modale Formeln

```

1 theory HOMLinHOL imports Main
2 begin
3 —<Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4 nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5 declare[[syntax_ambiguity_warning=false]]
6 —<Type i is associated with possible worlds and type e with entities:>
7 typedecl i —<Possible worlds>
8 typedecl e —<Individuals/entities>
9 type_synonym  $\sigma$  = "i $\Rightarrow$ bool" —<World-lifted propositions>
10 type_synonym  $\tau$  = "e $\Rightarrow$  $\sigma$ " —<Modal properties>
11 consts R::"i $\Rightarrow$ i $\Rightarrow$ bool" ("_r_") —<Accessibility relation between worlds>
12 axiomatization where
13   Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
14 —<Logical connectives (operating on truth-sets):>
15 abbreviation Mbot:: $\sigma$  ("⊥") where " $\bot \equiv \lambda w. \text{False}$ "
16 abbreviation Mtop:: $\sigma$  ("⊤") where " $\top \equiv \lambda w. \text{True}$ "
17 abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬") [52]53) where " $\neg \varphi \equiv \lambda w. \neg(\varphi w)$ "
18 abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∧" 50) where " $\varphi \wedge \psi \equiv \lambda w. \varphi w \wedge \psi w$ "
19 abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨" 49) where " $\varphi \vee \psi \equiv \lambda w. \varphi w \vee \psi w$ "
20 abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃" 48) where " $\varphi \supset \psi \equiv \lambda w. \varphi w \longrightarrow \psi w$ "
21 abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "↔" 47) where " $\varphi \leftrightarrow \psi \equiv \lambda w. \varphi w \longleftrightarrow \psi w$ "
22 abbreviation Mbox::" $\sigma \Rightarrow \sigma$ " ("□") [54]55) where " $\Box \varphi \equiv \lambda w. \forall v. w r v \longrightarrow \varphi v$ "
23 abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇") [54]55) where " $\Diamond \varphi \equiv \lambda w. \exists v. w r v \wedge \varphi v$ "
24 abbreviation Mprimeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("=_") where " $x = y \equiv \lambda w. x = y$ "
25 abbreviation Mprimneg::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≠") where " $x \neq y \equiv \lambda w. x \neq y$ "
26 abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬") where " $\sim \Phi \equiv \lambda x. \lambda w. \neg \Phi x w$ "
27 abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where " $\Phi . \Psi \equiv \lambda x. \lambda w. \Phi x w \wedge \Psi x w$ "
28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where " $\varphi \vee^e \psi \equiv (\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ "
29 —<Possibilist quantifiers (polymorphic)>
30 abbreviation Mallposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀") where " $\forall \Phi \equiv \lambda w. \forall x. \Phi x w$ "
31 abbreviation Mallpossb (binder "∀" [8]9) where " $\forall x. \varphi(x) \equiv \forall \varphi$ "
32 abbreviation Mexiposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃") where " $\exists \Phi \equiv \lambda w. \exists x. \Phi x w$ "
33 abbreviation Mexipossb (binder "∃" [8]9) where " $\exists x. \varphi(x) \equiv \exists \varphi$ "
34 —<Actualist quantifiers (for individuals/entities)>
35 consts existsAt::" $e \Rightarrow \sigma$ " ("@")
36 abbreviation Mallact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀E") where " $\forall^E \Phi \equiv \lambda w. \forall x. x @ w \longrightarrow \Phi x w$ "
37 abbreviation Mallactb (binder "∀E" [8]9) where " $\forall^E x. \varphi(x) \equiv \forall^E \varphi$ "
38 abbreviation Mexiact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃E") where " $\exists^E \Phi \equiv \lambda w. \exists x. x @ w \wedge \Phi x w$ "
39 abbreviation Mexiactb (binder "∃E" [8]9) where " $\exists^E x. \varphi(x) \equiv \exists^E \varphi$ "
40 —<Leibniz equality (polymorphic):>
41 abbreviation Mleibeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≡") where " $x \equiv y \equiv \forall P. P x \supset P y$ "
42 —<Meta-logical predicate for global validity:>
43 abbreviation Mvalid::" $\sigma \Rightarrow \text{bool}$ " ("⊢") where " $\vdash \psi \equiv \forall w. \psi w$ "
44 end

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKEy methodology

HOML in HOL



Ein Mathematiker würde nur diese Gleichungen betrachten

```

1 theory HOMLinHOL imports Main
2 begin
3 —<Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4 nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5 declare[[syntax_ambiguity_warning=false]]
6 —<Type i is associated with possible worlds and type e with entities:>
7 typedecl i —<Possible worlds>
8 typedecl e —<Individuals/entities>
9 type_synonym  $\sigma$  = "i $\Rightarrow$ bool" —<World-lifted propositions>
10 type_synonym  $\tau$  = "e $\Rightarrow$  $\sigma$ " —<Modal properties>
11 consts R::"i $\Rightarrow$ i $\Rightarrow$ bool" ("_r_") —<Accessibility relation between worlds>
12 axiomatization where
13   Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
14 —<Logical connectives (operating on truth-sets):>
15 abbreviation Mbot:: $\sigma$  ("⊥") where
16 abbreviation Mtop:: $\sigma$  ("⊤") where
17 abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬_") [52]53) where
18 abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∧" 50) where
19 abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨" 49) where
20 abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃" 48) where
21 abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "↔" 47) where
22 abbreviation Mbox::" $\sigma \Rightarrow \sigma$ " ("□_") [54]55) where
23 abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇_") [54]55) where
24 abbreviation Mprimeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("=_") where
25 abbreviation Mprimneg::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≠_") where
26 abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬_") where
27 abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where
28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where
29 —<Possibilist quantifiers (polymorphic)>
30 abbreviation Mallposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀") where
31 abbreviation Mallpossb (binder "∀" [8]9) where
32 abbreviation Mexiposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃") where
33 abbreviation Mexipossb (binder "∃" [8]9) where
34 —<Actualist quantifiers (for individuals/entities)>
35 consts existsAt::" $e \Rightarrow \sigma$ " ("@_")
36 abbreviation Mallact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀E") where
37 abbreviation Mallactb (binder "∀E" [8]9) where
38 abbreviation Mexiact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃E") where
39 abbreviation Mexiactb (binder "∃E" [8]9) where
40 —<Leibniz equality (polymorphic):>
41 abbreviation Mleibeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≡_") where
42 —<Meta-logical predicate for global validity:>
43 abbreviation Mvalid::" $\sigma \Rightarrow \text{bool}$ " ("⊢_") where
44 end

```

```

"⊥ ≡ λw. False"
"⊤ ≡ λw. True"
"¬φ ≡ λw. ¬(φ w)"
"φ ∧ ψ ≡ λw. φ w ∧ ψ w"
"φ ∨ ψ ≡ λw. φ w ∨ ψ w"
"φ ⊃ ψ ≡ λw. φ w ⟶ ψ w"
"φ ↔ ψ ≡ λw. φ w ⟷ ψ w"
"□φ ≡ λw. ∀v. w r v ⟶ φ v"
"◇φ ≡ λw. ∃v. w r v ∧ φ v"
"x = y ≡ λw. x = y"
"x ≠ y ≡ λw. x ≠ y"
"¬Φ ≡ λx. λw. ¬Φ x w"
"Φ.Ψ ≡ λx. λw. Φ x w ∧ Ψ x w"
"φ ∨e ψ ≡ (φ ∨ ψ) ∧ ¬(φ ∧ ψ)"

"∀Φ ≡ λw. ∀x. Φ x w"
"∀x. φ(x) ≡ ∀φ"
"∃Φ ≡ λw. ∃x. Φ x w"
"∃x. φ(x) ≡ ∃φ"

"∀EΦ ≡ λw. ∀x. x @ w ⟶ Φ x w"
"∀Ex. φ(x) ≡ ∀Eφ"
"∃EΦ ≡ λw. ∃x. x @ w ∧ Φ x w"
"∃Ex. φ(x) ≡ ∃Eφ"

"x ≡ y ≡ ∀P. P x ⊃ P y"

"⊢ψ ≡ ∀w. ψ w"

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKey methodology

HOML in HOL: Tests

S5-Axiome: gültig wie erwartet

```
1 theory TestsHOML imports HOMLinHOL
2 begin
3   <Test for S5 modal logic>
4   lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5   lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6   lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7   lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8   lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
```



Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: Tests

S5-Axiome: gültig wie erwartet
Barcan-Formeln: nicht gültig für 'Varying-Dom.'-Quant.
Barcan-Formeln: gültig für 'Constant-Dom.'-Quant.

```
1 theory TestsHOML imports HOLinHOL
2 begin
3 -<Test for S5 modal logic>
4 lemma axM: "[□φ ⊃ φ]" using Rrefl by blast
5 lemma axD: "[□φ ⊃ ◇φ]" using Rrefl by blast
6 lemma axB: "[φ ⊃ □◇φ]" using Rsymm by blast
7 lemma ax4: "[□φ ⊃ □□φ]" using Rtrans by blast
8 lemma ax5: "[◇φ ⊃ □◇φ]" using Rsymm Rtrans by blast
9 -<Test for Barcan and converse Barcan formula:>
10 lemma BarcanAct: "[ (∀x. □(φ x)) ⊃ □(∀x. (φ x)) ]" nitpick oops -<Countermodel found>
11 lemma ConvBarcanAct: "[ □(∀x. (φ x)) ⊃ (∀x. □(φ x)) ]" nitpick oops -<Countermodel found>
12 lemma BarcanPoss: "[ (∀x. □(φ x)) ⊃ □(∀x. φ x) ]" by blast
13 lemma ConvBarcanPoss: "[ □(∀x. (φ x)) ⊃ (∀x. □(φ x)) ]" by blast
```



Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: Tests

- S5-Axiome: gültig wie erwartet
- Barcan-Formeln: nicht gültig für 'Varying-Dom.'-Quant.
- Barcan-Formeln: gültig für 'Constant-Dom.'-Quant.
- Hilbert-Axiome und -Regeln: gültig wie erwartet

```
1 theory TestsHOML imports HOLinHOL
2 begin
3 <Test for S5 modal logic>
4 lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5 lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6 lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7 lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8 lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
9 <Test for Barcan and converse Barcan formula>
10 lemma BarcanAct: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. (\varphi x))$ ]" nitpick oops <Countermodel found>
11 lemma ConvBarcanAct: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" nitpick oops <Countermodel found>
12 lemma BarcanPoss: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. \varphi x)$ ]" by blast
13 lemma ConvBarcanPoss: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" by blast
14 <A simple Hilbert system for classical propositional logic is derived>
15 lemma Hilbert_A1: "[ $A \supset (B \supset A)$ ]" by blast
16 lemma Hilbert_A2: "[ $(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))$ ]" by blast
17 lemma Hilbert_MP: assumes "[A]" and "[A  $\supset$  B]" shows "[B]" using assms by blast
```



Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: Tests

- S5-Axiome: gültig wie erwartet
- Barcan-Formeln: nicht gültig für 'Varying-Dom.'-Quant.
- Barcan-Formeln: gültig für 'Constant-Dom.'-Quant.
- Hilbert-Axiome und -Regeln: gültig wie erwartet
- Existenzieller Import: ja, für 'Constant-Domain'-Quant.

```
1 theory TestsHOML imports HOLinHOL
2 begin
3   <Test for S5 modal logic>
4   lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5   lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6   lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7   lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8   lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
9   <Test for Barcan and converse Barcan formula>
10  lemma BarcanAct: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. (\varphi x))$ ]" nitpick oops <Countermodel found>
11  lemma ConvBarcanAct: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" nitpick oops <Countermodel found>
12  lemma BarcanPoss: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. \varphi x)$ ]" by blast
13  lemma ConvBarcanPoss: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" by blast
14  <A simple Hilbert system for classical propositional logic is derived>
15  lemma Hilbert_A1: "[ $A \supset (B \supset A)$ ]" by blast
16  lemma Hilbert_A2: "[ $(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))$ ]" by blast
17  lemma Hilbert_MP: assumes "[A]" and "[A  $\supset$  B]" shows "[B]" using assms by blast
18  <We have a polymorphic possibilist quantifier for which existential import holds>
19  lemma Quant_1: assumes "[A]" shows "[ $\forall x::'a. A$ ]" using assms by auto
20  <Existential import holds for possibilist quantifiers>
21  lemma ExImPossibilist1: "[ $\exists x::e. x = x$ ]" by blast
22  lemma ExImPossibilist2: "[ $\exists x::e. x \equiv x$ ]" by blast
23  lemma ExImPossibilist3: "[ $\exists x::e. x = t$ ]" by blast
24  lemma ExImPossibilist4: "[ $\exists x::'a. x \equiv t::'a$ ]" by blast
25  lemma ExImPossibilist: "[ $\exists x::'a. T$ ]" by blast
```



Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: Tests

- S5-Axiome: gültig wie erwartet
- Barcan-Formeln: nicht gültig für 'Varying-Dom.'-Quant.
- Barcan-Formeln: gültig für 'Constant-Dom.'-Quant.
- Hilbert-Axiome und -Regeln: gültig wie erwartet
- Existenzieller Import: ja, für 'Constant-Domain'-Quant.
- Existenzieller Import: nein, für 'Varying-Domain'-Quant.

```
1 theory TestsHOML imports HOLinHOL
2 begin
3 <Test for S5 modal logic>
4 lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5 lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6 lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7 lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8 lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
9 <Test for Barcan and converse Barcan formula>
10 lemma BarcanAct: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. (\varphi x))$ ]" nitpick oops <Countermodel found>
11 lemma ConvBarcanAct: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" nitpick oops <Countermodel found>
12 lemma BarcanPoss: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. \varphi x)$ ]" by blast
13 lemma ConvBarcanPoss: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" by blast
14 <A simple Hilbert system for classical propositional logic is derived>
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17 lemma Hilbert_MP: assumes "[A]" and "[A  $\supset$  B]" shows "[B]" using assms by blast
18 <We have a polymorphic possibilist quantifier for which existential import holds>
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20 <Existential import holds for possibilist quantifiers>
21 lemma ExImPossibilist1: "[ $\exists x::e. x = x$ ]" by blast
22 lemma ExImPossibilist2: "[ $\exists x::e. x \equiv x$ ]" by blast
23 lemma ExImPossibilist3: "[ $\exists x::e. x = t$ ]" by blast
24 lemma ExImPossibilist4: "[ $\exists x::'a. x \equiv t::'a$ ]" by blast
25 lemma ExImPossibilist: "[ $\exists x::'a. T$ ]" by blast
26 <We have an actualist quantifier for individuals for which existential import does not hold>
27 lemma Quant_2: assumes "[A]" shows "[ $\forall x::e. A$ ]" using assms by auto
28 <Existential import does not hold for our actualist quantifiers (for individuals)>
29 lemma ExImActualist1: "[ $\exists x::e. x = x$ ]" nitpick[card=1] oops <Countermodel found>
30 lemma ExImActualist2: "[ $\exists x::e. x \equiv x$ ]" nitpick[card=1] oops <Countermodel found>
31 lemma ExImActualist3: "[ $\exists x::e. x = t$ ]" nitpick[card=1] oops <Countermodel found>
32 lemma ExImActualist: "[ $\exists x::e. T$ ]" nitpick[card=1] oops <Countermodel found>
```



Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

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- S5-Axiome: gültig wie erwartet
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- Hilbert-Axiome und -Regeln: gültig wie erwartet
- Existenzieller Import: ja, für 'Constant-Domain'-Quant.
- Existenzieller Import: nein, für 'Varying-Domain'-Quant.
- Eigenschaften der Gleichheit: gültig wie erwartet
(außer für boolesche Extensionalität)

```
1 theory TestsHOML imports HOLinHOL
2 begin
3 <Test for S5 modal logic>
4 lemma axM: "[⊡φ ⊃ φ]" using Rrefl by blast
5 lemma axD: "[⊡φ ⊃ ⊡φ]" using Rrefl by blast
6 lemma axB: "[φ ⊃ ⊡φ]" using Rsymm by blast
7 lemma ax4: "[⊡φ ⊃ ⊡⊡φ]" using Rtrans by blast
8 lemma ax5: "[⊡φ ⊃ ⊡⊡φ]" using Rsymm Rtrans by blast
9 <Test for Barcan and converse Barcan formula:>
10 lemma BarcanAct: "[⊡(∀x.⊡(φ x)) ⊃ ⊡(∀x.(φ x))]" nitpick oops <Countermodel found>
11 lemma ConvBarcanAct: "[⊡(∀x.(φ x)) ⊃ ⊡(∀x.⊡(φ x))]" nitpick oops <Countermodel found>
12 lemma BarcanPoss: "[⊡(∀x.⊡(φ x)) ⊃ ⊡(∀x. φ x)]" by blast
13 lemma ConvBarcanPoss: "[⊡(∀x.(φ x)) ⊃ ⊡(∀x.⊡(φ x))]" by blast
14 <A simple Hilbert system for classical propositional logic is derived>
15 lemma Hilbert_A1: "[A ⊃ (B ⊃ A)]" by blast
16 lemma Hilbert_A2: "[⊡(A ⊃ (B ⊃ C)) ⊃ ⊡((A ⊃ B) ⊃ (A ⊃ C))]" by blast
17 lemma Hilbert_MP: assumes "[A]" and "[A ⊃ B]" shows "[B]" using assms by blast
18 <We have a polymorphic possibilist quantifier for which existential import holds>
19 lemma Quant_1: assumes "[A]" shows "[∀x::'a. A]" using assms by auto
20 <Existential import holds for possibilist quantifiers>
21 lemma ExImPossibilist1: "[⊡∃x::e. x = x]" by blast
22 lemma ExImPossibilist2: "[⊡∃x::e. x ≡ x]" by blast
23 lemma ExImPossibilist3: "[⊡∃x::e. x = t]" by blast
24 lemma ExImPossibilist4: "[⊡∃x::'a. x ≡ t::'a]" by blast
25 lemma ExImPossibilist: "[⊡∃x::'a. T]" by blast
26 <We have an actualist quantifier for individuals for which existential import does not hold>
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30 lemma ExImActualist2: "[⊡∃x::e. x ≡ x]" nitpick[card=1] oops <Countermodel found>
31 lemma ExImActualist3: "[⊡∃x::e. x = t]" nitpick[card=1] oops <Countermodel found>
32 lemma ExImActualist: "[⊡∃x::e. T]" nitpick[card=1] oops <Countermodel found>
33 <Properties of the embedded primitive equality, which coincides with Leibniz equality>
34 lemma EqRefI: "[x = x]" by blast
35 lemma EqSym: "[x = y] ↔ [y = x]" by blast
36 lemma EqTrans: "[⊡(x = y) ∧ ⊡(y = z)] ⊃ ⊡(x = z)" by blast
37 lemma EQCong: "[⊡(x = y) ⊃ ⊡(φ x) = ⊡(φ y)]" by blast
38 lemma EQFuncExt: "[⊡(φ = ψ) ⊃ ⊡(∀x. (⊡(φ x) = ⊡(ψ x)))]" by blast
39 lemma EQBoolExt1: "[⊡(φ = ψ) ⊃ ⊡(φ ↔ ψ)]" by blast
40 lemma EQBoolExt2: "[⊡(φ ↔ ψ) ⊃ ⊡(φ = ψ)]" nitpick[card=2] oops <Countermodel found>
41 lemma EQBoolExt3: "[⊡(φ ↔ ψ)] → ⊡(φ = ψ)" by blast
42 lemma EqPrimLeib: "[⊡(x = y) ↔ ⊡(x ≡ y)]" by auto
```



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- Hilbert-Axiome und -Regeln: gültig wie erwartet
- Existenzieller Import: ja, für 'Constant-Domain'-Quant.
- Existenzieller Import: nein, für 'Varying-Domain'-Quant.
- Eigenschaften der Gleichheit: gültig wie erwartet
(außer für boolesche Extensionalität)
- Volle Komprehension: gültig wie erwartet

```
1 theory TestsHOML imports HOLinHOL
2 begin
3 <Test for S5 modal logic>
4 lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5 lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6 lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7 lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8 lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
9 <Test for Barcan and converse Barcan formula:>
10 lemma BarcanAct: "[ $(\forall x. \Box (\varphi x)) \supset \Box (\forall x. (\varphi x))$ ]" nitpick oops <Countermodel found>
11 lemma ConvBarcanAct: "[ $\Box (\forall x. (\varphi x)) \supset (\forall x. \Box (\varphi x))$ ]" nitpick oops <Countermodel found>
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14 <A simple Hilbert system for classical propositional logic is derived>
15 lemma Hilbert_A1: "[ $A \supset (B \supset A)$ ]" by blast
16 lemma Hilbert_A2: "[ $(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))$ ]" by blast
17 lemma Hilbert_MP: assumes "[A]" and "[A  $\supset$  B]" shows "[B]" using assms by blast
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24 lemma ExImPossibilist4: "[ $\exists x::a. x \equiv t::a$ ]" by blast
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33 <Properties of the embedded primitive equality, which coincides with Leibniz equality>
34 lemma EqRef1: "[ $x = x$ ]" by blast
35 lemma EqSym: "[ $(x = y) \leftrightarrow (y = x)$ ]" by blast
36 lemma EqTrans: "[ $((x = y) \wedge (y = z)) \supset (x = z)$ ]" by blast
37 lemma EQCong: "[ $(x = y) \supset ((\varphi x) = (\varphi y))$ ]" by blast
38 lemma EQFuncExt: "[ $(\varphi = \psi) \supset (\forall x. ((\varphi x) = (\psi x)))$ ]" by blast
39 lemma EQBoolExt1: "[ $(\varphi = \psi) \supset (\varphi \leftrightarrow \psi)$ ]" by blast
40 lemma EQBoolExt2: "[ $(\varphi \leftrightarrow \psi) \supset (\varphi = \psi)$ ]" nitpick[card=2] oops <Countermodel found>
41 lemma EQBoolExt3: "[ $(\varphi \leftrightarrow \psi) \longrightarrow [(\varphi = \psi)]$ ]" by blast
42 lemma EqPrimLeib: "[ $(x = y) \leftrightarrow (x \equiv y)$ ]" by auto
43 <Comprehension is natively supported in HOL (due to  $\lambda$ -abstraction)>
44 lemma Comprehension1: "[ $\exists \varphi. \forall x. (\varphi x) \leftrightarrow A$ ]" by force
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- Eigenschaften der Gleichheit: gültig wie erwartet
(außer für boolesche Extensionalität)
- Volle Komprehension: gültig wie erwartet
- Modaler Kollaps: nicht gültig

```
1 theory TestsHOML imports HOLinHOL
2 begin
3   <Test for S5 modal logic>
4   lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5   lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6   lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7   lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8   lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
9   <Test for Barcan and converse Barcan formula>
10  lemma BarcanAct: "[ $(\forall x. \Box (\varphi x)) \supset \Box (\forall x. (\varphi x))$ ]" nitpick oops <Countermodel found>
11  lemma ConvBarcanAct: "[ $\Box (\forall x. (\varphi x)) \supset (\forall x. \Box (\varphi x))$ ]" nitpick oops <Countermodel found>
12  lemma BarcanPoss: "[ $(\forall x. \Diamond (\varphi x)) \supset \Diamond (\forall x. \varphi x)$ ]" by blast
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15  lemma Hilbert_A1: "[ $A \supset (B \supset A)$ ]" by blast
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17  lemma Hilbert_MP: assumes "[A]" and "[A  $\supset$  B]" shows "[B]" using assms by blast
18  <We have a polymorphic possibilist quantifier for which existential import holds>
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34  lemma EqRef1: "[ $x = x$ ]" by blast
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37  lemma EQCong: "[ $(x = y) \supset ((\varphi x) = (\varphi y))$ ]" by blast
38  lemma EQFuncExt: "[ $(\varphi = \psi) \supset (\forall x. ((\varphi x) = (\psi x)))$ ]" by blast
39  lemma EQBoolExt1: "[ $(\varphi = \psi) \supset (\varphi \leftrightarrow \psi)$ ]" by blast
40  lemma EQBoolExt2: "[ $(\varphi \leftrightarrow \psi) \supset (\varphi = \psi)$ ]" nitpick[card=2] oops <Countermodel found>
41  lemma EQBoolExt3: "[ $(\varphi \leftrightarrow \psi) \longrightarrow [(\varphi = \psi)]$ ]" by blast
42  lemma EqPrimLeib: "[ $(x = y) \leftrightarrow (x \equiv y)$ ]" by auto
43  <Comprehension is natively supported in HOL (due to  $\lambda$ -abstraction)>
44  lemma Comprehension1: "[ $\exists \varphi. \forall x. (\varphi x) \leftrightarrow A$ ]" by force
45  lemma Comprehension2: "[ $\exists \varphi. \forall x. (\varphi x) \leftrightarrow (A x)$ ]" by force
46  lemma Comprehension3: "[ $\exists \varphi. \forall x y. (\varphi x y) \leftrightarrow (A x y)$ ]" by force
47  <Modal collapse does not hold>
48  lemma ModalCollapse: "[ $\forall \varphi. \varphi \supset \Box \varphi$ ]" nitpick[card=2] oops <Countermodel found>
```



Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

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- Eigenschaften der Gleichheit: gültig wie erwartet
(außer für boolesche Extensionalität)
- Volle Komprehension: gültig wie erwartet
- Modaler Kollaps: nicht gültig
- universelle Eigenschaft = Selbstidentität
- Absurde/leere Eigenschaft = Selbstdifferenz

```
1 theory TestsHOML imports HOLinHOL
2 begin
3   <Test for S5 modal logic>
4   lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5   lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
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7   lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8   lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
9   <Test for Barcan and converse Barcan formula>
10  lemma BarcanAct: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. (\varphi x))$ ]" nitpick oops <Countermodel found>
11  lemma ConvBarcanAct: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" nitpick oops <Countermodel found>
12  lemma BarcanPoss: "[ $(\forall x. \Diamond(\varphi x)) \supset \Diamond(\forall x. (\varphi x))$ ]" by blast
13  lemma ConvBarcanPoss: "[ $\Diamond(\forall x. (\varphi x)) \supset (\forall x. \Diamond(\varphi x))$ ]" by blast
14  <A simple Hilbert system for classical propositional logic is derived>
15  lemma Hilbert_A1: "[ $A \supset (B \supset A)$ ]" by blast
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24  lemma ExImPossibilist4: "[ $\exists x::a. x \equiv t::a$ ]" by blast
25  lemma ExImPossibilist: "[ $\exists x::a. T$ ]" by blast
26  <We have an actualist quantifier for individuals for which existential import does not hold>
27  lemma Quant_2: assumes "[A]" shows "[ $\forall x::e. A$ ]" using assms by auto
28  <Existential import does not hold for our actualist quantifiers (for individuals)>
29  lemma ExImActualist1: "[ $\exists x::e. x = x$ ]" nitpick[card=1] oops <Countermodel found>
30  lemma ExImActualist2: "[ $\exists x::e. x \equiv x$ ]" nitpick[card=1] oops <Countermodel found>
31  lemma ExImActualist3: "[ $\exists x::e. x = t$ ]" nitpick[card=1] oops <Countermodel found>
32  lemma ExImActualist: "[ $\exists x::e. T$ ]" nitpick[card=1] oops <Countermodel found>
33  <Properties of the embedded primitive equality, which coincides with Leibniz equality>
34  lemma EqRefI: "[ $x = x$ ]" by blast
35  lemma EqSym: "[ $(x = y) \leftrightarrow (y = x)$ ]" by blast
36  lemma EqTrans: "[ $((x = y) \wedge (y = z)) \supset (x = z)$ ]" by blast
37  lemma EQCong: "[ $(x = y) \supset ((\varphi x) = (\varphi y))$ ]" by blast
38  lemma EQFuncExt: "[ $(\varphi = \psi) \supset (\forall x. ((\varphi x) = (\psi x)))$ ]" by blast
39  lemma EQBoolExt1: "[ $(\varphi = \psi) \supset (\varphi \leftrightarrow \psi)$ ]" by blast
40  lemma EQBoolExt2: "[ $(\varphi \leftrightarrow \psi) \supset (\varphi = \psi)$ ]" nitpick[card=2] oops <Countermodel found>
41  lemma EQBoolExt3: "[ $(\varphi \leftrightarrow \psi) \longrightarrow [(\varphi = \psi)]$ ]" by blast
42  lemma EqPrimLeib: "[ $(x = y) \leftrightarrow (x \equiv y)$ ]" by auto
43  <Comprehension is natively supported in HOL (due to  $\lambda$ -abstraction)>
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46  lemma Comprehension3: "[ $\exists \varphi. \forall x y. (\varphi x y) \leftrightarrow (A x y)$ ]" by force
47  <Modal collapse does not hold>
48  lemma ModalCollapse: "[ $\forall \varphi. \varphi \supset \Box \varphi$ ]" nitpick[card=2] oops <Countermodel found>
49  <Empty property and self-difference>
50  lemma TruePropertyAndSelfIdentity: "[ $(\lambda x::e. T) = (\lambda x. x = x)$ ]" by blast
51  lemma EmptyPropertyAndSelfDifference: "[ $(\lambda x::e. \bot) = (\lambda x. x \neq x)$ ]" by blast
52  lemma EmptyProperty2: "[ $\exists x. \varphi x \longrightarrow \varphi \neq (\lambda x::e. \bot)$ ]" by blast
53  lemma EmptyProperty3: "[ $\exists \exists x. \varphi x \longrightarrow \varphi \neq (\lambda x::e. \bot)$ ]" by blast
54
55 end
```



Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: modaler (Ultra-)Filter

Auf HOML-
Ebene geliftete
Mengen-
operationen

```
1 theory ModalFilter imports HOMLinHOL
2 begin
3 type_synonym  $\tau = "e \Rightarrow \sigma"$ 
4 abbreviation Element:: $\tau \Rightarrow (\tau \Rightarrow \sigma) \Rightarrow \sigma$  (infix  $\in$  90) where
5 abbreviation EmptySet:: $\tau$  (" $\emptyset$ ") where
6 abbreviation UniversalSet:: $\tau$  (" $U$ ") where
7 abbreviation Subset:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq$  80) where
8 abbreviation SubsetE:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq^E$  80) where
9 abbreviation Intersection:: $\tau \Rightarrow \tau \Rightarrow \tau$  (infix  $\sqcap$  91) where
10 abbreviation Inverse:: $\tau \Rightarrow \tau$  (" $^{-1}$ ") where
11 abbreviation "Filter  $\Phi \equiv U \in \Phi \wedge \neg(\emptyset \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \varphi \subseteq^E \psi \supset \psi \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \psi \in \Phi \supset \varphi \sqcap \psi \in \Phi)"$ 
12 abbreviation "UFilter  $\Phi \equiv \text{Filter } \Phi \wedge (\forall \varphi. \varphi \in \Phi \vee (-^1 \varphi) \in \Phi)"$ 
13 abbreviation "FilterP  $\Phi \equiv U \in \Phi \wedge \neg(\emptyset \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \varphi \subseteq \psi \supset \psi \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \psi \in \Phi \supset \varphi \sqcap \psi \in \Phi)"$ 
14 abbreviation "UFilterP  $\Phi \equiv \text{FilterP } \Phi \wedge (\forall \varphi. \varphi \in \Phi \vee (-^1 \varphi) \in \Phi)"$ 
15 end
```

$$\begin{aligned} \varphi \in S &\equiv S \varphi \\ \emptyset &\equiv \lambda x. \perp \\ U &\equiv \lambda x. \top \\ \varphi \subseteq \psi &\equiv \forall x. ((\varphi x) \supset (\psi x)) \\ \varphi \subseteq^E \psi &\equiv \forall^E x. ((\varphi x) \supset (\psi x)) \\ \varphi \sqcap \psi &\equiv \lambda x. ((\varphi x) \wedge (\psi x)) \\ {}^{-1}\psi &\equiv \lambda x. \neg(\psi x) \end{aligned}$$

Fig. 5 Set filter and ultrafilter formalized for our modal logic setting

- Auf HOML-Ebene geliftete Begriffe von Filter und Ultrafilter (für Varying-/Constant-Dom.-Quant.)
- enthält die Allmenge, aber nicht die leere Menge
 - abgeschlossen unter Obermengen und Schnitt
 - maximal (nur für Ultrafilter)

Onto-Logischer Beweis Feb. 10, 1970

$P(\varphi)$ φ is positive (i.e. $\varphi \in P$)

At. 1 $P(\varphi) \cdot P(\psi) \supset P(\varphi \cdot \psi)$ At. 2 $P(\varphi) \cdot \neg P(\psi) \supset P(\varphi \cdot \neg \psi)$

[1] $G(x) \equiv (\varphi) [P(\varphi) \supset \varphi(x)]$ (Good)

[2] $\varphi \text{ Ess } x \equiv (\psi) [\psi(x) \supset N(\exists y) [\varphi(y) \supset \psi(y)]]$ (Essence of x)

$p \supset_N q = N(p \supset q)$ Necessity

At. 2 $P(\varphi) \supset NP(\varphi)$ because it follows from the nature of the property

Th. $G(x) \supset G \text{ Ess } x$

Df. $E(x) \equiv (\varphi) [\varphi \text{ Ess } x \supset N(\exists x) \varphi(x)]$ necessary Existence

Ax 3 $P(E)$

Th. $G(x) \supset N(\exists y) G(y)$

hence $(\exists x) G(x) \supset N(\exists y) G(y)$

" $M(\exists x) G(x) \supset MN(\exists y) G(y)$ $M = possibility$

" $\supset N(\exists y) G(y)$

any two sentences of x are nec. equivalent

exclusive or $\cdot$ and for any number of humanoids

$M(\exists x) G(x)$ means ^{the system of} all pos. props. is compatible

This is true because of:

At. 4: $P(\varphi) \cdot \varphi \supset_N \psi \supset P(\psi)$ which impl.

~~hence~~ $\begin{cases} x=x & \text{is positive} \\ x \neq x & \text{is negative} \end{cases}$

But if a system S of pos. props. were inconsistent it would mean that the same prop. S (which is positive) would be $x \neq x$

Positive means positive in the moral aesthetic sense (independently of the accidental structure of the world). Only when the ax. true. It may also mean "attribution" as opposed to "privation" (or containing privation). This interpretation provides proof

If φ is positive: $(x) N \sim \varphi(x)$ - Otherwise $\varphi(x) \supset_N x \neq x$

hence $x \neq x$ positive $\supset_N x = x$ necessity At. 4

on the other hand $\varphi(x) \supset_N x \neq x$

hence $x \neq x$ is negative

x i.e. the normal form in terms of elem. prop. contains a member without negation.

Formalisierung von Gödels Manuskript von 1970 (Scriptum-Begriff der Essenz: Gödel1)

— Widerspruch —

Gödel 1

Positive properties: P

Axiom Ax1: $P\varphi \wedge P\psi \supset P(\varphi . \psi)$

Axiom Ax2a: $P\varphi \vee^e P\sim\varphi$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi . \psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
```

P: Positive Eigenschaften
(Intensionen)

Ax1:

Die Konjunktion zweier positiver
Eigenschaften ist positiv.

Ax2a:

Entweder eine Eigenschaft oder ihr
Komplement ist positiv.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Definition of God-like: $G x \equiv \forall \varphi. P \varphi \supset \varphi x$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall \varphi. P \varphi \supset \varphi x$ "
```

G(ottähnlich):
Eine Entität x ist gottähnlich
gdw.
es alle positiven Eigenschaften besitzt.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Definition of Essence: $\varphi \text{ Ess. } x \equiv \forall \psi. \psi x \supset (\varphi \supset_N \psi)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ _") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi \text{ Ess. } x \equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
```

Ess(enz):

Eine Eigenschaft φ ist Essenz von x
gdw.

alle Eigenschaften von x notwendigerweise
durch φ impliziert sind.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Axiom Ax2b: $P\varphi \supset \Box P\varphi$ and $\neg P\varphi \supset \Box(\neg P\varphi)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi \supset$  P ( $\varphi . \psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi \vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv \forall\varphi. P\varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P\varphi$ ]"
```

Ax2b:

Positive Eigenschaften sind
notwendigerweise positiv.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Axiom Ax2b: $P \varphi \supset \Box P \varphi$ and $\neg P \varphi \supset \Box(\neg P \varphi)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8 definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
```

Lemma:
Nicht-positive Eigenschaften sind
notwendigerweise nicht-positiv.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Theorem Th1: $G x \supset G \text{Ess. } x$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10  lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
```

Theorem 1:
Gottähnlichkeit ist Essenz
jeder gottähnlichen Entität.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Definition of Necessary Existence: $E x \equiv \forall \varphi. \varphi \text{ Ess. } x \supset \Box(\exists^E x. \varphi x)$

Axiom Ax3: $P E$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ _") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi \text{ Ess. } x \equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall\varphi. \varphi \text{ Ess. } x \supset \Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
```

E(xistenz):

x hat die Eigenschaft der Existenz
gdw.

alle Essenzen von x notwendigerweise
exemplifiziert sind.

Ax3:

E(xistenz) ist eine positive Eigenschaft.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Theorem Th2: $G x \supset \Box(\exists^E y. G y)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi x$ "
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8 definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9 axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10 lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset \Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
```

Theorem 2:

Wenn x gottähnlich ist, dann ist die Eigenschaft der Gottähnlichkeit notwendigerweise exemplifiziert.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Theorem Th3: $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$

Logik KB (Rsymm) hier benötigt

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
```

Theorem:

Die mögliche Existenz einer gottähnlichen Entität impliziert die notwendige Existenz einer gottähnlichen Entität.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Axiom Ax4: $P \varphi \wedge (\varphi \supset_N \psi) \supset P \psi$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi \wedge P \psi \supset P (\varphi \cdot \psi)$ ]"
5   axiomatization where Ax2a: "[P  $\varphi \vee^e P \sim \varphi$ ]"
6   definition God ("G") where "G x  $\equiv \forall \varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ _") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi \text{ Ess. } x \equiv \forall \psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall \varphi. \varphi \text{ Ess. } x \supset \Box(\exists^e x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^e x. G x) \supset \Diamond \Box(\exists^e y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi \wedge (\varphi \supset_N \psi) \supset P \psi$ ]"
```

Ax4:

Eigenschaften, die notwendigerweise durch eine positive Eigenschaft impliziert sind, sind positiv.

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Kein Modell gefunden!

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   abbreviation PropertyInclusion ("_ $\supset_N$ _") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10  lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^e x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset$   $\Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^e x. G x)$   $\supset$   $\Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^e x. G x)$   $\supset$   $\Box(\exists^e y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^e x. G x)$   $\supset$   $\Diamond\Box(\exists^e y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^e x. G x)$   $\supset$   $\Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23  lemma True nitpick[satisfy] oops —<No model found>
```

Konsistenzprüfung mit Nitpick:
... kein Modell gefunden ...

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Widerspruch entdeckt
durch Theorembeweiser

Der Widerspruch folgt über das
„Lemma der leeren Essenz“:

$$(\lambda x. \perp) \text{ Ess. } x$$



Anpassung des Essenz-Begriffs?

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4   axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5   axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6   definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7   abbreviation PropertyInclusion ("_⊃N_") where "φ ⊃N ψ ≡ □(∀ey. φ y ⊃ ψ y)"
8   definition Essence ("_Ess._") where "φ Ess. x ≡ ∀ψ. ψ x ⊃ (φ ⊃N ψ)"
9   axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10  lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃ex. φ x)"
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x ⊃ □(∃ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[◇(∃ex. G x) ⊃ □(∃ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ (∃ex. G x) ⊃ □(∃ey. G y)]" using Th2 by blast
18    have 2: "[ ◇(∃ex. G x) ⊃ ◇□(∃ey. G y)]" using 1 by blast
19    have 3: "[ ◇(∃ex. G x) ⊃ □(∃ey. G y)]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23  lemma True nitpick[satisfy] oops —<No model found>
24  lemma EmptyEssL: "[ (λy. ⊥) Ess. x]" using Essence_def by auto
```

Fig. 6 Gödel’s axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Widerspruch entdeckt
durch Theorembeweiser

Der Widerspruch folgt über das
„Lemma der leeren Essenz“:

$(\lambda x. \perp) \text{ Ess. } x$



Anpassung des Essenz-Begriffs?

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi$  x"
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi \text{ Ess. } x \equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall\varphi. \varphi \text{ Ess. } x \supset \Box(\exists^e x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^e x. G x) \supset \Diamond\Box(\exists^e y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi \wedge (\varphi \supset_N \psi) \supset P \psi$ ]"
23  lemma True nitpick[satisfy] oops —<No model found>
24  lemma EmptyEssL: "[ $(\lambda y. \perp) \text{ Ess. } x$ ]" using Essence_def by auto
25  theorem Inconsistency: False sledgehammer(Ax2a Ax3 Ax4 EmptyEssL NecExist_def) —<Proof found>
26  proof -
27    have 1: "[ $\neg(P (\lambda x. \perp))$ ]" using Ax2a Ax4 by blast
28    have 2: "[P ( $\lambda x. (\lambda y. \perp) \text{ Ess. } x \supset \Box(\exists^e z. (\lambda y. \perp) z)$ )]" using Ax3 Ax4 NecExist_def by smt
29    have 3: "[P ( $\lambda x. \Box(\exists^e z. (\lambda x. \perp) z)$ )]" using 2 EmptyEssL by simp
30    have 4: "[P ( $\lambda x. \Box \perp$ )]" using 3 by auto
31    have 5: "[P ( $\lambda x. \perp$ )]" using 4 Ax2a Ax4 by smt
32    have 6: "[ $\perp$ ]" using 1 5 by blast
33    thus ?thesis by blast
34  qed
35 end
```

Widerspruch
... bewiesen ...

Fig. 6 Gödel’s axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1 — Ergebnisse

(Positive) Eigenschaften als intensionale Prädikate

‘ φ impliziert ψ notwendigerweise’ definiert als $\Box \forall^E x . \varphi x \supset \psi x$

Ax1: Die Konjunktion zweier positiver Eigenschaften ist positiv.

Ax2a: Entweder eine Eigenschaft oder ihr Komplement ist positiv.

G(ottähnlich): x ist gottähnlich gdw. es alle positiven Eigenschaften besitzt.

Ess(enz): φ ist Essenz von x gdw. alle Eigenschaften von x notwendigerweise durch φ impliziert sind.

Ax2b: Positive Eigenschaften sind notwendigerweise positiv.

Theorem 1: Gottähnlichkeit ist die Essenz jeder gottähnlichen Entität.

E(xistenz): x hat Existenz gdw. alle Essenzen von x notwendigerweise exemplifiziert sind.

Ax3: E(xistenz) ist eine positive Eigenschaft.

Theorem 2: Wenn x gottähnlich ist, dann ist Gottähnlichkeit notwendigerweise exemplifiziert.

Theorem 3: Die mögliche Existenz einer gottähnlichen Entität impliziert die notwendige Existenz einer gottähnlichen Entität.

Ax4: Eigenschaften, die notwendigerweise durch eine positive Eigenschaft impliziert sind, sind positiv.

Diese Postulate sind **widersprüchlich**
(bereits in **Modallogik K**)

Formalisierung von Gödels Manuskript von 1970 (angepasster Essenz-Begriff und generalisierten Axioms Ax1Gen: Gödel2)

Ontologischer Beweis Feb. 10, 1970

$P(\varphi)$ φ is positive ($\varphi \in P$)

Ax 1 $P(\varphi) \cdot P(\psi) \supset P(\varphi \cdot \psi)$ Ax 2 $P(\varphi) \cdot \neg P(\psi) \supset P(\varphi \cdot \neg \psi)$

[1] $G(x) \equiv (\varphi) [P(\varphi) \supset \varphi(x)]$ (Gödel)

[2] $\varphi \text{ Ess. } x \equiv (\psi) [\psi(x) \supset N(\psi) [\varphi(y) \supset \psi(y)]]$ (Essence of x)

$p \supset_N q = N(p \supset q)$ Necessity

Ax 2 $P(\varphi) \supset N P(\varphi)$
 $\neg P(\varphi) \supset N \neg P(\varphi)$ } because it follows from the nature of the property

Th. $G(x) \supset G \text{ Ess. } x$

Df. $E(x) \equiv (\varphi) [\varphi \text{ Ess. } x \supset N \exists x \varphi(x)]$ necessary Existence

Ax 3 $P(E)$

Th. $G(x) \supset N(\exists y) G(y)$

hence $(\exists x) G(x) \supset N(\exists y) G(y)$

" $M(\exists x) G(x) \supset M N(\exists y) G(y)$

" $\supset N(\exists y) G(y)$ $M = possibility$

any two instances of x are nec. equivalent
exclusive or * and for any number of humanoids

$M(\exists x) G(x)$ means ^{the system of} all pos. props. as com-
patible This is true because of:

Ax 4: $P(\varphi) \cdot \varphi \supset_N \psi \supset P(\psi)$ which impl.

~~True~~ $\begin{cases} x=x & \text{is positive} \\ x \neq x & \text{is negative} \end{cases}$

But if a system S of pos. props. were incomp. it would mean that the sum prop. S (which is positive) would be $x \neq x$

Positive means positive in the moral aesth. sense (independently of the accidental structure of the world). Only when the act is true. It also means "attribution" as opposed to "privation" (or containing privation). This interprets the proof of φ privation: $(x) N \neg \varphi(x)$ - Obvious: $\varphi(x) \supset_N x \neq x$ hence $x \neq x$ positive not $x=x$ neg. violating Ax. or the equiv. of pos. $M(\exists x) G(x)$

Th. x i.e. the normal form in terms of elem. prop. contains a member without negation.

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— Beweis in KB —

$\varphi \supset \Box \Diamond \varphi$

Formalisierung von Gödels Manuskript von 1970 (angepasster Essenz- Begriff und generalisierten Axioms Ax1Gen: Gödel2)

— Beweis in KB —

$$\varphi \supset \Box \Diamond \varphi$$

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4 axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5 axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6 definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7 abbreviation PropertyInclusion ("⊃N") where "φ ⊃N ψ ≡ □(∀Ey. φ y ⊃ ψ y)"
8 (**) definition Essence ("Ess.") where "φ Ess. x ≡ φ x ∧ ∀ψ. ψ x ⊃ (φ ⊃N ψ)"
9 axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10 lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃Ex. φ x)"
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x ⊃ □(∃Ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using Th2 by blast
18   have 2: "[◇(∃Ex. G x) ⊃ ◇□(∃Ey. G y)]" using 1 by blast
19   have 3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23 lemma True nitpick[satisfy,card=1,eval="[P (λx. ⊥)]"] oops —<One model found of cardinality one>
24 abbreviation "PosProps φ ≡ ∀ψ. φ ψ ⊃ P ψ"
25 abbreviation "ConjOfPropsFrom φ φ ≡ □(∀Ez. φ z ↔ (∀ψ. φ ψ ⊃ ψ z))"
26 axiomatization where Ax1Gen: "[ (PosProps φ ∧ ConjOfPropsFrom φ φ) ⊃ P φ ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[◇(∃Ex. G x)]" using Ax2a Ax4 L by blast
29 theorem Th5: "[□(∃Ex. G x)]" using Th3 Th4 by blast
30 lemma MC: "[φ ⊃ □φ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: "[∀x.(G x w → (∀Z. Z x ⊃ □(∀Ez. G z ⊃ Z z)) w)]" using Ax2a Ax2b God_def by smt
33   have 2: "[∃x. G x w → ((Q ⊃ □(∀Ez. G z ⊃ Q)) w)]" using 1 by force
34   have 3: "[Q ⊃ □Q] w" using 2 Th5 Rsymm by blast}
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P (λx. ⊤) ∧ P (λx. x = x)]" using Ax2a Ax4 by blast
38 lemma NegProps: "[¬P (λx. ⊥) ∧ ¬P (λx. x ≠ x)]" using Ax2a Ax4 by blast
39 lemma UniqueEss1: "[φ Ess. x ∧ ψ Ess. x ⊃ □(∀Ey. φ y ↔ ψ y)]" using Essence_def by smt
40 lemma UniqueEss2: "[φ Ess. x ∧ ψ Ess. x ⊃ □(φ ≡ ψ)]" nitpick[card i=1] oops —<Countermodel found>
41 lemma UniqueEss3: "[φ Ess. x ⊃ □(∀Ey. φ y ⊃ y ≡ x)]" using Essence_def MC by auto
42 lemma Monotheism: "[G x ∧ G y ⊃ x ≡ y]" using Ax2a God_def by smt
43 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P (λx. ⊥)]"] oops —<One model found of cardinality one>
46 end
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2 — Ergebnisse

‘ φ impliziert ψ notwendigerweise’ gdw. $\Box \forall^E x . \varphi x \supset \psi x$

Ax1Gen: Die Konjunktion **beliebiger** positiver Eigenschaften ist positiv.

Ax2a: Entweder eine Eigenschaft oder ihr Komplement ist positiv.

G(ottähnlich): x ist gottähnlich gdw. es alle positiven Eigenschaften besitzt.

Ess(enz): φ ist Essenz von x gdw. alle Eigenschaften von x notwendigerweise durch φ impliziert sind **und x die Eigenschaft φ hat.**

Ax2b: Positive Eigenschaften sind notwendigerweise positiv.

Theorem 1: Gottähnlichkeit ist die Essenz jeder gottähnlichen Entität.

E(xistenz): x hat Existenz gdw. alle Essenzen von x notwendigerweise exemplifiziert sind.

Ax3: E(xistenz) ist eine positive Eigenschaft.

Theorem 2: Wenn x gottähnlich ist, dann ist Gottähnlichkeit notwendigerweise exemplifiziert.

Theorem 3: Die mögliche Existenz einer gottähnlichen Entität impliziert die notwendige Existenz einer gottähnlichen Entität.

Ax4: Eigenschaften, die notwendigerweise durch eine positive Eigenschaft impliziert sind, sind positiv.

Lemma L: Gottähnlichkeit ist eine positive Eigenschaft.

Theorem 4: Eine gottähnliche Entität existiert möglicherweise.

Theorem 5: Eine gottähnliche Entität existiert notwendigerweise.

Verallgemeinerung von Ax1: **Ax1Gen**

• and for any number of minimally

Axiom Ax1Gen: $(PosProps \Phi \wedge ConjOfPropsFrom \varphi \Phi) \supset P \varphi$

abbreviation "PosProps $\Phi \equiv \forall \varphi. \Phi \varphi \supset P \varphi$ "

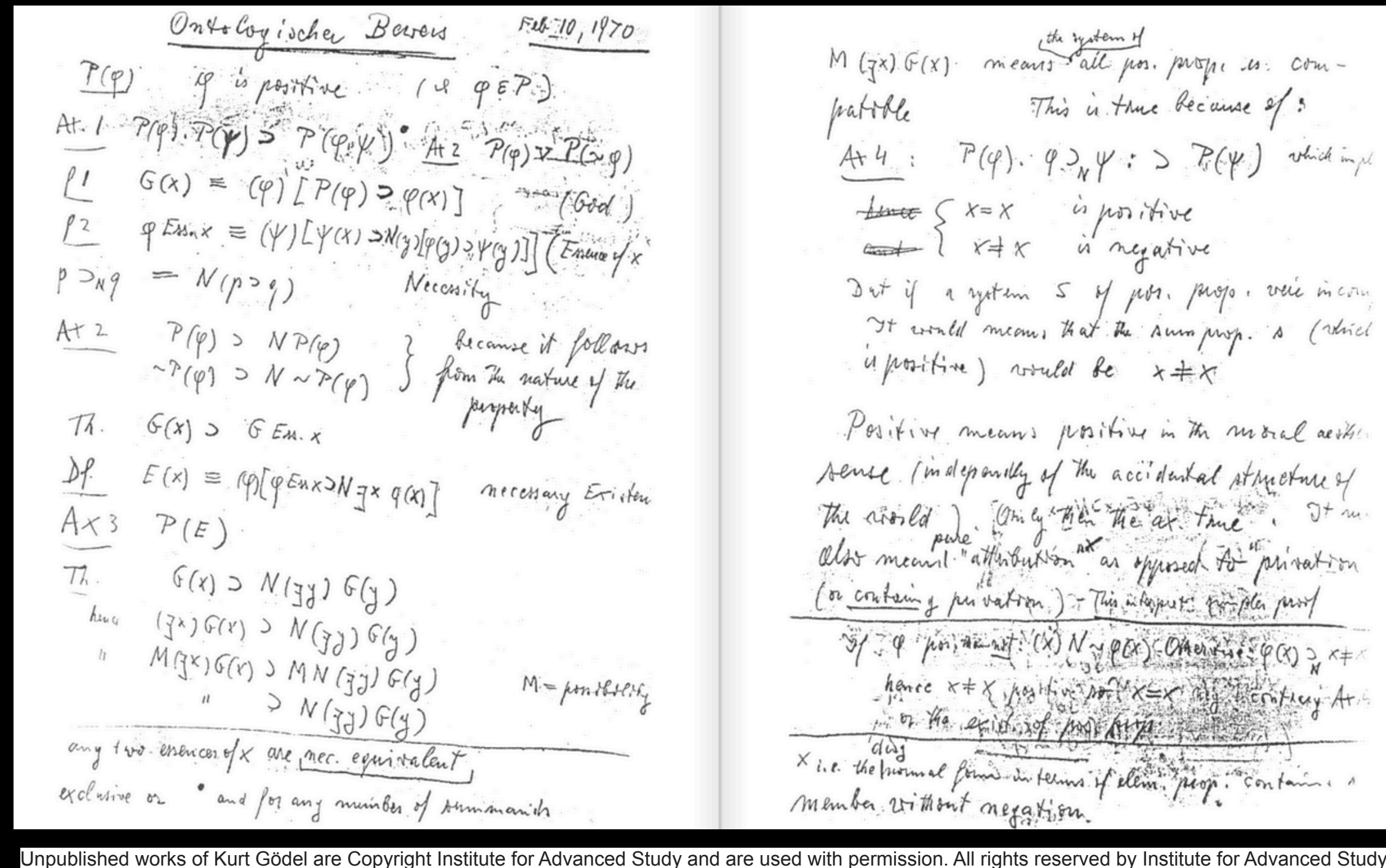
abbreviation "ConjOfPropsFrom $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall \psi. \Phi \psi \supset \psi z))$ "

axiomatization where Ax1Gen: " $[(PosProps \Phi \wedge ConjOfPropsFrom \varphi \Phi) \supset P \varphi]$ "

Ax1Gen modellierbar in Logik 3.Ordnung

- Widerspruchsfrei
- Gültig bereits in Modallogik **KB**
- **Modaler Kollaps** folgt
- **Monotheismus** folgt
- **P** ist ein **Ultrafilter**
- Selbstidentität: positive Eigenschaft
- Selbstdifferenz: nicht-positive Eigens.

Formalisierung von Gödels Manuskript von 1970 (angepasste Eigenschaftsinklusion und generalisierte Axioms Ax1 Gen: Gödel3)



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— Beweis in KB —

$$\varphi \supset \Box \Diamond \varphi$$

Hinweis: Dies ist eine neue Variante

Formalisierung von Gödels Manuskript von 1970 (angepasste Eigenschaftsinkklusion und generalisierte Axioms Ax1Gen: Gödel3)

— Beweis in KB —

$$\varphi \supset \Box \Diamond \varphi$$

```

1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4   axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5   axiomatization where Ax2a: "[P φ ∨ P ~φ]"
6   definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7   (**) abbreviation PropertyInclusion ("⊃N") where "φ ⊃N ψ ≡ □ φ ≠ (λx. ⊥) ∧ ∀ey. φ y ⊃ ψ y)"
8   definition Essence ("_Ess._") where "φ Ess. x ≡ ∀ψ. ψ x ⊃ (φ ⊃N ψ)"
9   axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10  lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃Ex. φ x)"
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x ⊃ □(∃Ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using Th2 by blast
18    have 2: "[◇(∃Ex. G x) ⊃ ◇□(∃Ey. G y)]" using 1 by blast
19    have 3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23  lemma True nitpick[satisfy,card=1,eval="[P (λx. ⊥)]"] oops —<Two models found of cardinality one>
24  abbreviation "PosProps Φ ≡ ∀φ. Φ φ ⊃ P φ"
25  abbreviation "ConjOfPropsFrom φ Φ ≡ □(∀Ez. φ z ↔ (∀ψ. Φ ψ ⊃ ψ z))"
26  axiomatization where Ax1Gen: "[ (PosProps Φ ∧ ConjOfPropsFrom φ Φ) ⊃ P φ ]"
27  lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
28  theorem Th4: "[◇(∃Ex. G x)]" sledgehammer[timeout=200](Ax2a L Ax1Gen) sorry —<Proof found>
29  theorem Th5: "[□(∃Ex. G x)]" using Th3 Th4 by blast
30  lemma MC: "[φ ⊃ □φ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31  proof - {fix w fix Q
32    have 1: "∀x.(G x w → (∀Z. Z x ⊃ □(∀Ez. G z ⊃ Z z)) w)" using Ax2a Ax2b God_def by smt
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34    have 3: "(Q ⊃ □Q) w" using 2 Th5 Rsymm by blast}
35  thus ?thesis by auto
36  qed
37  lemma PosProps: "[P (λx. T) ∧ P(λx. x = x)]" using Ax2a Ax4 L Th4 by smt
38  lemma NegProps: "[¬P(λx. ⊥) ∧ ¬P(λx. x ≠ x)]" using Ax2a Ax4 L Th4 by smt
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42  lemma Monotheism: "[G x ∧ G y ⊃ x ≡ y]" using Ax2a God_def by smt
43  lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44  lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45  lemma True nitpick[satisfy,card=1,eval="[P (λx. T)]"] oops —<One model found of cardinality one>
46 end

```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 3 — Ergebnisse

“Teufel = ... (leere Menge)”

(Max III, S. 47)

“Bemerkung: Die leere Menge (der leere Raum) ist ein Amphibium zwischen Nichts und Etwas.”
(Max III, S. 85)

‘ φ impliziert ψ notwendigerweise’ gdw. $\Box \forall^E x . \varphi \neq \lambda x . \perp \wedge \varphi x \supset \psi x$

Ax1Gen: Die Konjunktion **beliebiger** positiver Eigenschaften ist positiv.

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Ax3: E(xistenz) ist eine positive Eigenschaft.

Theorem 2: Wenn x gottähnlich ist, dann ist Gottähnlichkeit notwendigerweise exemplifiziert.

Theorem 3: Die mögliche Existenz einer gottähnlichen Entität impliziert die notwendige Existenz einer gottähnlichen Entität.

Ax4: Eigenschaften, die notwendigerweise durch eine positive Eigenschaft impliziert sind, sind positiv.

Lemma L: Gottähnlichkeit ist eine positive Eigenschaft.

Theorem 4: Eine gottähnliche Entität existiert möglicherweise.

Theorem 5: Eine gottähnliche Entität existiert notwendigerweise.

← Die leere Menge impliziert nun keine anderen Eigenschaften mehr !

- Widerspruchsfrei
- Gültig bereits in Modallogik **KB**
- **Modaler Kollaps** folgt
- **Monotheismus** folgt
- **P** ist ein **Ultrafilter**
- Selbstidentität: positive Eigenschaft
- Selbstdifferenz: nicht-positive Eigens.

Gödel 3 — Ergebnisse

“Teufel = ... (leere Menge)” (Max III, S. 47)
“Bemerkung: Die leere Menge (der leere Raum) ist ein Amphibium zwischen Nichts und Etwas.” (Max III, S. 85)

‘ φ impliziert ψ notwendigerweise’ gdw. $\Box \forall^E x . \varphi \neq \lambda x . \perp \wedge \varphi x \supset \psi x$

Die leere Menge impliziert nun keine anderen Eigenschaften mehr !

Simplified Variant of Gödel's Ontological Argument

explored through systematic simplification experiments with a modern higher-order theorem proving technology

Simplified Axioms

- A1' (C1): Self-difference (empty property) is not a positive property.
- A2' (C2): A property is positive if it is entailed by a positive property.
- A3: Being Godlike is a positive property.*
- (Modal logic K: Necessitation rule N, distribution axiom K)

Single Definition/Shorthand:

D1: An entity x is Godlike if it possesses all positive properties.

Simplified Argument (mainly non-modal, except for last step; note also that D1 is not used)

- L1: The existence of a non-exemplified positive property implies that self-difference (resp. the empty property) is positive.
(from A2', since such a non-exemplified positive property would entail self-difference (p entails q: for all x, p(x) implies q(x))
- L2: A non-exemplified positive property does not exist.
(from A1' and the contrapositive of L1)
- T1': Positive properties are exemplified.
(reformulation of L2)
- T4: There exists a Godlike being.
(from A3, T1')
- T3: Necessarily, there exists a Godlike being.
(from T4, by necessitation)

* Alternatively, we may postulate A3': The conjunction of any collection of positive properties is positive; then we prove A3 from A3' and D1.

Further Findings

- Consistent (model finder Nitpick: one world, one Godlike being)
- Modal collapse not implied (countermodel by Nitpick: two possible worlds i1 and i2 and with one Godlike entity e; i1 and i2 are reachable from i2, but only i1 can be reached from i1; there is a non-positive property ϕ which holds for e in i2 but not in i1)
- But also the attempt to prove that the existence of a Godlike being is possible fails (countermodel by Nitpick: single world with no other world reachable)
- Way out: we simply postulate the additional modal axiom T (what necessarily holds, that holds; resp. what holds is possible)

Trivialisierende Simplifikationen — wie ich sie auch selbst publiziert habe — werden dadurch eliminiert.

Symbolic AI and Gödel's Ontological Argument. Zygon 57(4):953–962.
Doi: 10.1111/zygo.12830

A Simplified Variant of Gödel's Ontological Argument. In: A. Vestrucci (ed.), Beyond Babel: Religions and Linguistic Pluralism. Doi: 10.1007/978-3-031-42127-3_19

Gödel 2

Zusammenfassung

- benötigt das verallgemeinerte Axiom **Ax1Gen**
- modifiziert: **Essenz**
- kein existenzieller Import
- widerspruchsfrei
- mögliche und notwendige Existenz von G
- verifiziert bereits in Logik **KB**
- abgeleitete Ergebnisse:
 - **modaler Kollaps**
 - **Monotheismus**
 - **P ist ein Ultrafilter**

Gödel 3

Zusammenfassung

- benötigt das verallgemeinerte Axiom **Ax1Gen**
- modifiziert: **Eigenschaftsinklusion**
- kein existenzieller Import
- widerspruchsfrei
- mögliche und notwendige Existenz von G
- verifiziert bereits in Logik **KB**
- abgeleitete Ergebnisse:
 - **modaler Kollaps**
 - **Monotheismus**
 - **P ist ein Ultrafilter**

Wechsel der Quantoren für Entitäten von
‘Varying-Domain’ zu ‘Constant-Domain’? —> Ergebnisse bestätigt

Und der Teufel?

“Nämlich, um ein Ding {a} zu erkennen, muss ich haben: 1.) gewisse fundamentale Zuordnungen gewisser empirischer Dinge (vielleicht beginnen mit **Gott = Allmenge***; **Teufel = ein unwiderlegbar falsches System*** (oder leere Menge); ...” (Max III, S. 47)

Bem<erkung> (Theol<ogie>): Die vollkommen klare Erkenntnis des {vollkommen} Bösen, dass er seine Absicht nie erreichen kann, bedeutet seinen Tod. (Max IX, S. 36)

Und der Teufel? — Variante 1 (Dank an Chad Brown)

```
1 theory ThereIsNoEvil1 imports GoedelVariantHOML2
2 begin
3 definition Evil ("Evil") where "Evil x  $\equiv \forall \varphi. \neg P \varphi \supset \varphi x$ "
4 theorem NecNoEvil: "[ $\Box(\neg(\exists^E x. \text{Evil } x))$ ]" sledgehammer(Ax2a Ax4 Evil_def) —<Proof found>
5 proof -
6   have "[ $\neg P(\lambda y. \perp)$ ]" using Ax2a Ax4 by blast
7   hence "[ $(\forall^E x. \text{Evil } x \supset (\lambda y. \perp) x)$ ]" using Evil_def by auto
8   hence "[ $(\forall^E x. \text{Evil } x \supset \perp)$ ]" by auto
9   hence "[ $(\exists^E x. \text{Evil } x) \supset \perp$ ]" by auto
10  hence "[ $\neg(\exists^E x. \text{Evil } x)$ ]" by blast
11  thus ?thesis by blast
12 qed
13 end
```

Gödel 2

x ist Teufel gdw.
x alle nicht-positiven
Eigenschaften besitzt.

+

Axiome wie bisher



Notwendigerweise
existiert kein Teufel.

Fig. 10 Importing Gödel's modified axioms from Fig. 7 we can prove that necessarily there exists no entity that possesses all non-positive (=negative) properties

```
1 theory ThereIsNoEvil2 imports GoedelVariantHOML3
2 begin
3 definition Evil ("Evil") where "Evil x  $\equiv \forall \varphi. \neg P \varphi \supset \varphi x$ "
4 theorem NecNoEvil: "[ $\Box(\neg(\exists^E x. \text{Evil } x))$ ]" sledgehammer(Ax1Gen Ax2a Evil_def) oops —<Proof found>
5 end
```

Gödel 3

Bestätigt auch für
Gödel 3

Fig. 11 Importing Gödel's modified axioms from Fig. 8 we can prove that necessarily there exists no entity that possesses all non-positive (=negative) properties

Und der Teufel? — Variante 2 (Scott)

```
1 theory EvilDerivable imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 definition Evil ("Evil") where "Evil x  $\equiv$   $\forall \varphi. \neg P \varphi \supset \varphi x$ "
5 definition Essence ("_Ess.") where " $\varphi$  Ess. x  $\equiv$   $\varphi x \wedge (\forall \psi. \psi x \supset \Box(\forall^E y. \varphi y \supset \psi y))$ "
6 definition NecExist ("E") where "E x  $\equiv$   $\forall \varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
7 axiomatization where A1: " $\neg P \varphi \leftrightarrow P \neg \varphi$ "
8 axiomatization where A2: " $\neg P \varphi \wedge \Box(\forall^E y. \varphi y \supset \psi y) \supset \neg P \psi$ "
9 axiomatization where A4: " $\neg P$  Evil"
10 axiomatization where A3: " $\neg P \varphi \supset \Box(\neg P \varphi)$ "
11 axiomatization where A5: " $\neg P E$ "
12 lemma True nitpick[satisfy, card i=1, eval="P ( $\lambda x. \bot$ )", eval="P ( $\lambda x. T$ )"] oops —<Model found>
13 theorem T1: " $\neg P \varphi \supset \Diamond(\exists^E x. \varphi x)$ " using A1 A2 by blast
14 theorem T2: " $\Diamond(\exists^E x. \text{Evil } x)$ " using A4 T1 by blast
15 theorem T3: " $\text{Evil } x \supset \text{Evil Ess. } x$ " using A1 A3 Essence_def Evil_def by (smt (verit, best))
16 theorem T4: " $\Diamond(\exists^E x. \text{Evil } x) \supset \Box(\exists^E y. \text{Evil } y)$ " using A5 Evil_def NecExist_def Rsymm T3 by smt
17 theorem T5: " $\Box(\exists^E x. \text{Evil } x)$ " using T2 T4 by presburger
18 lemma MC: " $\varphi \supset \Box \varphi$ " sledgehammer(A1 A3 T5 Evil_def Rsymm) oops —<proof found>
19 lemma PosProps: " $P (\lambda x. \bot) \wedge P (\lambda x. x \neq x)$ " using A1 A2 by blast
20 lemma NegProps: " $\neg P (\lambda x. T) \wedge \neg P (\lambda x. x = x)$ " using A1 A2 by blast
21 lemma UniqueEss1: " $\varphi$  Ess. x  $\wedge \psi$  Ess. x  $\supset \Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ " using Essence_def by smt
22 lemma UniqueEss2: " $\varphi$  Ess. x  $\wedge \psi$  Ess. x  $\supset \Box(\varphi = \psi)$ " nitpick[card i=2] oops —<Countermodel found>
23 lemma Monoevilism: " $\text{Evil } x \wedge \text{Evil } y \supset x \equiv y$ " using A1 Evil_def by smt
24 lemma Filter: " $\text{Filter } (\lambda \varphi. \neg P \varphi)$ " using A1 Evil_def Rsymm T1 T5 by (smt (verit, best))
25 lemma UltraFilter: " $\text{UFilter } (\lambda \varphi. \neg P \varphi)$ " using Filter A1 by blast
26 end
```

Fig. 20 The necessary existence of an Evil-like entity proved from (controversially) modified assumptions

x ist Teufel gdw. x alle nicht-positiven Eigenschaften besitzt
+
Spiegelung aller Axiome (Scott)
für nicht-positive Eigenschaften



Notwendigerweise existiert
ein Teufel

Aber: Die Eigenschaften dieses Teufels sind absolut identisch mit den
Eigenschaften von Gödels gottähnlicher Entität!

Gödel's Scriptum-Variante (1970) und mathematischer Realismus

Gott = (maximal) unendlich/unfassbar?

Bem<erkung> (Phil<osophie>): Wenn man die Objekte der Mat<hematik> als durch den Geist konstruiert ansieht, bringt man notwendig ein zeitlich<es> Element hinein.

(Max IX, S. 45)

“Gott = Allmenge”

(Max III, S. 47)

Bem<erkung> (Gr<undlagen>): Ist das universelle ‚Alle‘ begrifflich früher [d. h. zum Verständnis notwendig] des Speziellen, zum Beispiel: alle Zahlen? Es scheint nicht, sondern man lernt das Alle sogar an endlichen Beispielen [z. B.: alle Spielsteine auf diesem Tisch] und erreicht das universelle <Alle> überhaupt niemals.

(Max IX, S. 22)

Gödel 2 — als Startpunkt ...

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter ChurchTTinHOML
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where
5 axiomatization where
6 definition God ("G") where
7 abbreviation PropertyInclusion (" $\supset_N$ ") where
8 definition Essence ("_Ess.") where
9 axiomatization where
10 lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]"
11 theorem Th1: "[ $G x \supset G \text{Ess. } x$ ]"
12 definition NecExist ("E") where
13 axiomatization where
14 theorem Th2: "[ $G x \supset \Box(\exists^E y. G y)$ ]"
15 theorem Th3: "[ $\Diamond(\exists^F x. G x) \supset \Box(\exists^F y. G y)$ ]"
16 axiomatization where
17 lemma True nitpick[satisfy,card=1,eval="[ $P(\lambda x. \bot)$ ]" ] oops —<One model found of cardinality one>
18 abbreviation CollOfPosProps  $\Phi = \forall \varphi. \Phi \varphi \supset P \varphi$ 
19 abbreviation ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^F z. \varphi z \leftrightarrow (\forall \psi. \Phi \psi \supset \psi z))$ 
20 axiomatization where
21 lemma L: "[ $P G$ ]"
22 theorem Th4: "[ $\Diamond(\exists^F x. G x)$ ]"
23 theorem Th5: "[ $\Box(\exists^F x. G x)$ ]"
24 lemma MC: "[ $\varphi \supset \Box \varphi$ ]" —<Proof found with sledgehammer>
25 proof - {fix w fix Q
26 have 1: " $\forall x. (G x w \longrightarrow (\forall z. z x \supset \Box(\forall^E z. G z \supset z z)) w)$ " using Ax2a Ax2b God_def by smt
27 have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^F z. G z \supset Q)) w)$ " using 1 by force
28 have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast
29 thus ?thesis by auto qed
30 lemma PosProps: "[ $P(\lambda x. \top) \wedge P(\lambda x. x = x)$ ]" using Ax2a Ax4 by blast
31 lemma NegProps: "[ $\neg P(\lambda x. \bot) \wedge \neg P(\lambda x. x \neq x)$ ]" using Ax2a Ax4 by blast
32 lemma UniqueEss1: "[ $\varphi \text{Ess. } x \wedge \psi \text{Ess. } x \supset \Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ ]" using Essence_def by smt
33 lemma UniqueEss2: "[ $\varphi \text{Ess. } x \wedge \psi \text{Ess. } x \supset \Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
34 lemma Monotheism: "[ $G x \wedge G y \supset x \equiv y$ ]" using Ax2a God_def by smt
35 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
36 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
37 lemma True nitpick[satisfy,user_axioms,card=1,eval="[ $P(\lambda x. \bot)$ ]" ] oops —<One model found of cardinality one>
```

Gödel 2 — plus mathematische Objekte

Annahme: Die Objekte der Mathematik
'existieren' (z.B. natürliche Zahlen n ,
Peano-Axiome).

Prädikate: $\lambda x. x = n$ und $\lambda x. x \neq n$

Implikationen für die Kardinalität der
positiven Eigenschaften P :

- P ist unendlich
- P ist überabzählbar unendlich

Behauptung (weitere Arbeit für Verifikation):

- wenn wir solche Eigenschaften wieder als existierend annehmen, dann zwingt das verallgemeinerte Axiom Ax1Gen die Kardinalität von P ins maximal Unendliche

On the maximality of positive properties and modal collapse in variants of Gödel's ontological proof of God

Cordelia Mühlenbeck, Christoph Benz Müller · *Logic and Logical Philosophy*, 1–21

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On Absolute Infinity and Modal Collapse in Gödel's Ontological Proof

Cordelia Mühlenbeck, Christoph Benz Müller · *Logic, Knowledge, and Tradition: Essays in Honor of Srećko Kovač* · In print

[BibTeX](#)

```
38 —<Assuming 2 different entities leads to 2 different positive properties, assuming 3 leads to 4, etc.>
39 lemma H: "[P (λe. T) ∧ (P (λe. e=a) ∨ P (λe. e≠a))]" using PosProps Ax2a by auto
40 lemma PP2: assumes "[∃(a::e) b. a ≠ b]" shows "[∃p q. p≠q ∧ P p ∧ P q]" by (smt (verit, del_insts) H assms)
41 lemma PP4: assumes "[∃(a::e) b c. a ≠ b ∧ a ≠ c ∧ b ≠ c]"
42   shows "[∃p q r s. p≠q ∧ p≠r ∧ p≠s ∧ q≠r ∧ q≠s ∧ r≠s ∧ P p ∧ P q ∧ P r ∧ P s]" by (smt (verit, del_insts) H assms)
43 —<Assuming infinitely many different entities leads to at least 2, 4, etc., many different positive properties>
44 lemma PP2': assumes "[Inf0]" shows "[∃p q. ((p≠q) ∧ P p ∧ P q)]" by (metis (lifting) Ax2a Inf0_def PosProps assms)
45 lemma PP4': assumes "[Inf0]" shows "[∃p q r s. p≠q ∧ p≠r ∧ p≠s ∧ q≠r ∧ q≠s ∧ r≠s ∧ P p ∧ P q ∧ P r ∧ P s]"
46 proof - have "[Inf0] ⇒ [∃p::e. ∃q::e. ∃r::e. p≠q ∧ p≠r ∧ q≠r]" unfolding Inf0_def apply simp by metis
47   thus ?thesis using PP4 assms by blast qed
48 —<Theorem MetaInf: infinitely many different entities lead to infinitely many positive properties: proof at meta-level.>
49 —<Various definitions and automatically proved lemmata are not shown here.> (*{ } { } *) [9 lines]
59 lemma MetaInf: "[Inf0] ⇒ infinite (UNIV::e set)" (*{ } { } *) [6 lines]
60 lemma InfPosSet_a: "infinite (UNIV::e set) ⇒ infinite (P w)" (*{ } { } *) [5 lines]
72 lemma InfPosSet_b: "[Inf0] ⇒ infinite (P w)" using InfPosSet_a MetaInf by blast (*{ } { } *) [6 lines]
79 theorem InfPosSet assumes "[Inf0]" shows "[¬(Finite P)]" by (metis Finite_def InfPosSet_b Nat_finite assms)
80 —<CantorArgument: Assuming infinitely many different entities leads to uncountably many different positive properties.>
81 definition S (f::nat⇒e) ≡ (λx::e. if x ∈ range f then f (inv f x) else x)
82 definition Enc (f::nat⇒e) s ≡ (λ(B::e⇒bool) (x::e) (u::i). x = f 0 ∨ (∃y. s y = x ∧ B y))
83 definition D F (f::nat⇒e) s (w::i) ≡ (λx::e. if F x (f 0) w then ¬F x (s x) w else F x (s x) w)
84 lemma Inf_inj: "[Inf0] ⇒ ∃f::nat⇒e. inj f" using MetaInf infinite_countable_subset by blast
85 lemma Enc0: "Enc f s B (f 0) w using Enc_def by simp
86 lemma EncS: "inj f ⇒ (Enc f (S f) B (S f y) w ↔ B y)"
87 by (smt Enc_def S_def inj_Suc inv_f_f inv_int_injective nat.distinct rangeI)
88 lemma L1: "inj (f::nat⇒e) ⇒ (¬(∃F. ∀p. P p ⊃ (∃e::e. F e = p))) w" by (smt Enc0 EncS Ax2a D_def)
89 theorem CantorArgument: assumes "[Inf0]" shows "[¬(∃F. ∀p. P p ⊃ (∃e::e. F e = p))]" using assms Inf_inj L1 by metis
90 end
```

—<Formalized notions: equipollence, cardinality, different notion of infinity>

22 abbreviation Ex1B (binder "∃₁" [8]9) where "∃₁x. Φ(x) ≡ (∃x. Φ(x) ∧ (∀z. Φ(z) ⊃ (z = x)))"

23 definition Equipollent ≡ (λp::'a⇒σ. λq::'b⇒σ. (∃s. (∀x. p x ⊃ q (s x)) ∧ (∀y. q y ⊃ (∃x. p x ∧ (y = s x)))))

24 definition Cardinality ≡ (λp::'b⇒σ. Equipollent p)

25 definition CardinalNumbers ≡ (λu::('b⇒σ)⇒σ. (∃p::('b⇒σ). (u = (Equipollent p))))

26 definition Inf0 ≡ ∃n::e⇒σ. ∀x y z. ((∃w. r x w) ∧ ¬(r x x) ∧ (r x y ⊃ (r y z ⊃ r x z)))

27 definition Inf1 ≡ ∃n::e⇒σ. ∃m::e⇒σ. (∀x. m x ⊃ n x) ∧ (∃u. n u ∧ ¬(m u)) ∧ (Equipollent n m)

28 definition Inf2 ≡ ∃n::e⇒σ. ∃u. ∃s. n u ∧ (∀x. n x ⊃ n (s x)) ∧ (∀x. n x ⊃ (s x)≠u) ∧ (∀x y. (n x ∧ n y ∧ s x=s y) ⊃ x=y)

29 —<Definitions/abbreviations for modalised natural numbers (modeled as analysable objects naturally contained in HOL)>

30 named theorems Nat

31 definition Zero ("0") where [Nat]: "0 ≡ ((=) (λx. 1))"

32 definition Succ ("S") where [Nat]: "S ≡ (λn. λp. (∃x. p x ∧ (n (λt. (t ≠ x) ∧ p t))))"

33 definition Naturals ("N") where [Nat]: "N ≡ (λn. (∀p. (p 0 ∧ (∀x. p x ⊃ p (S x)) ⊃ p n))"

34 definition Finite ("Finite") where [Nat]: "Finite ≡ (λp. (∃n. N n ∧ n p))"

Persönliches Fazit

Ontologischer Beweis Feb. 10, 1970

P(φ) φ is positive (i.e. φ ∈ P.)

At. 1 P(φ), P(ψ) ⊃ P(φ, ψ) At. 2 P(φ) ⊃ P(¬φ)

P1 G(x) ≡ (φ) [P(φ) ⊃ φ(x)] (God)

P2 φ Ess. x ≡ (ψ) [ψ(x) ⊃ N(y) [φ(y) ⊃ ψ(y)]] (Essence of x)

P ⊃_N q = N(p ⊃ q) Necessity

At. 2 P(φ) ⊃ NP(φ)
 ~P(φ) ⊃ N~P(φ) } because it follows from the nature of the property

Th. G(x) ⊃ G Ess. x

Def. E(x) ≡ (φ) [φ Ess. x ⊃ N ∃ x φ(x)] necessary Existence

Ax 3 P(E)

Th. G(x) ⊃ N(∃ y) G(y)

hence (∃ x) G(x) ⊃ N(∃ y) G(y)

" M(∃ x) G(x) ⊃ MN(∃ y) G(y) M = possibility

 " ⊃ N(∃ y) G(y)

any two essences of x are nec. equivalent,
 exclusive or and for any number of humanoids

member without negation.

Gödel's Scriptum-Variante von 1970

- mit modifizierter Eigenschaftsinklusion
- bei Beachtung seiner Fußnote

ist meine präferierte 'Lesart' seines modalen ontologischen Arguments.

Ich teile zudem die Auffassung von z.B. Kovač und Mühlenbeck (und anderen), dass Gödel den modalen Kollaps nicht nur akzeptiert, sondern beabsichtigt hat.

Die Untersuchung von Gödel's "Allmenge" (maximale Unendlichkeit) ist faszinierende weitere Arbeit (Typen- vs. Mengentheorie).

Literatur

Das ontologische Argument — Original & Schlüsselquellen

- Gödel, K. (2004/1970). **Appendix A: Notes in Kurt Gödel's Hand**. In: J.H. Sobel, *Logic and Theism*, S. 144–145. Cambridge University Press.
- Scott, D.S. (2004/1970). **Appendix B: Notes in Dana Scott's Hand**. In: J.H. Sobel, *Logic and Theism*, S. 145–146. Cambridge University Press. Doi: 10.1017/CBO9780511497988
- Sobel, J.H. (2004). **Logic and Theism: Arguments for and Against Beliefs in God**. Cambridge University Press. Doi: 10.1017/CBO9780511497988
- Adams, R.M. (1995). **Introductory Note to *1970**. In: Kurt Gödel: *Collected Works*, Bd. III, S. 388–402. Oxford University Press. Doi: 10.1093/oso/9780195072556.003.0017
- Kanckos, A. & Lethen, T. (2021). **The Development of Gödel's Ontological Proof**. *Review of Symbolic Logic* 14(4):1011–1029. Doi: 10.1017/S1755020319000479
- Wang, H. (1997). **A Logical Journey: From Gödel to Philosophy**. MIT Press. Doi: 10.7551/mitpress/4321.001.0001
- Gödel, K. (2019 ff.). **Philosophische Notizbücher / Philosophical Notebooks** (Hrsg. E.-M. Engelen, Übers. M. Carl). De Gruyter, Berlin/Boston. Doi: 10.1515/9783110583748 (Bd. 1); hier Max Phil 0, II–IX.

Literatur

Emendationen, Varianten & Kritik

- Anderson, C.A. (1990). **Some Emendations of Gödel's Ontological Proof**. Faith and Philosophy 7(3):291–303. Doi: 10.5840/faithphil19907325
- Anderson, C.A. & Gettings, M. (1996). **Gödel's Ontological Proof Revisited**. In: Gödel '96, Lecture Notes in Logic 6, S. 167–172. Cambridge University Press. Doi: 10.1017/9781316716939.011
- Hájek, P. (1996). **Magari and Others on Gödel's Ontological Proof**. In: Logic and Algebra, S. 125–135. Doi: 10.1201/9780203748671
- Hájek, P. (2002). **A New Small Emendation of Gödel's Ontological Proof**. Studia Logica 71(2):149–164. Doi: 10.1023/A:1016583920890
- Fitting, M. (2002). **Types, Tableaus, and Gödel's God**. Springer. Doi: 10.1007/978-94-010-0411-4
- Kovač, S. (2012). **Modal Collapse in Gödel's Ontological Proof**. In: Ontological Proofs Today, S. 323–343. Ontos. Doi: 10.1515/9783110325881.323
- Hazen, A.P. (1998). **On Gödel's Ontological Proof**. Australasian Journal of Philosophy 76(3):361–377. Doi: 10.1080/00048409812348501
- Bjørdal, F.A. (2018). **All Properties Are Divine or God Exists**. Logic and Logical Philosophy 27(3). Doi: 10.12775/LLP.2017.027

Literatur

Computergestützte Analyse von Gödels Argument

- Benz Müller, C. & Scott, D.S. (2025). **Notes on Gödel's and Scott's Variants of the Ontological Argument.** Monatshefte für Mathematik. Doi: 10.1007/s00605-025-02078-x
- Benz Müller, C. & Woltzenlogel Paleo, B. (2014). **Automating Gödel's Ontological Proof of God's Existence with Higher-order Automated Theorem Provers.** ECAI 2014. Doi: 10.3233/978-1-61499-419-0-93
- __ (2016). **The Inconsistency in Gödel's Ontological Argument: A Success Story for AI in Metaphysics.** IJCAI 2016 (<http://www.ijcai.org/Proceedings/16/Papers/137.pdf>)
- Benz Müller, C., Weber, L. & Woltzenlogel Paleo, B. (2017). **Computer-Assisted Analysis of the Anderson–Hájek Controversy.** Logica Universalis 11(1):139–151. Doi: 10.1007/s11787-017-0160-9
- Fuenmayor, D. & Benz Müller, C. (2017). **Automating Emendations of the Ontological Argument in Intensional Higher-Order Modal Logic.** KI 2017. Doi: 10.1007/978-3-319-67190-1_9
- Benz Müller, C. & Fuenmayor, D. (2020). **Computer-Supported Analysis of Positive Properties, Ultrafilters and Modal Collapse in Variants of Gödel's Ontological Argument.** Bulletin of the Section of Logic 49(2):127–148. Doi: 10.18778/0138-0680.2020.08
- Benz Müller, C. (2022). **Symbolic AI and Gödel's Ontological Argument.** Zygon 57(4):953–962. Doi: 10.1111/zygo.12830
- __ (2023). **A Simplified Variant of Gödel's Ontological Argument.** In: A. Vestrucci (ed.), Beyond Babel: Religions and Linguistic Pluralism. Springer. Doi: 10.1007/978-3-031-42127-3_19
- Vestrucci, A. & Benz Müller, C. (2024). **Reflections on the Ontological Proof of Kurt Gödel.** In: Divined Explanations, S. 255–285. Brill. Doi: 10.1163/9789004701908_013
- Mühlenbeck, C. & Benz Müller, C. (2026). **On the maximality of positive properties and modal collapse in variants of Gödel's ontological proof of God.** Logic and Logical Philosophy. Doi: 10.12775/LLP.2026.007

Methodische Grundlagen (Logik höherer Stufe, Modallogik, Werkzeuge)

- Church, A. (1940). **A Formulation of the Simple Theory of Types**. J. Symb. Logic 5:56–68. Doi: 10.2307/2266170
- Henkin, L. (1950). **Completeness in the Theory of Types**. J. Symb. Logic 15(2):81–91. Doi: 10.2307/2266967
- Benz Müller, C., Brown, C. & Kohlhasse, M. (2004). **Higher-Order Semantics and Extensionality**. J. Symb. Logic 69(4):1027–1088. Doi: 10.2178/jsl/1102022211
- Benz Müller, C. & Andrews, P. (2024). **Church's Type Theory**. Stanford Encyclopedia of Philosophy. (<https://plato.stanford.edu/entries/type-theory-church/>)
- Benz Müller, C. & Paulson, L. (2013). **Quantified Multimodal Logics in Simple Type Theory**. Logica Universalis 7(1):7–20. Doi: 10.1007/s11787-012-0052-y
- Scott, D.S. (1967). **Existence and Description in Formal Logic**. In: Bertrand Russell: Philosopher of the Century, S. 181–200. (Nachdruck in: Philosophical Applications of Free Logic, OUP 1991)
- Nipkow, T., Paulson, L.C. & Wenzel, M. (2002). **Isabelle/HOL — A Proof Assistant for Higher-Order Logic**. Springer, LNCS 2283. Doi: 10.1007/3-540-45949-9
- Blanchette, J.C. & Nipkow, T. (2010). **Nitpick: A Counterexample Generator for Higher-Order Logic**. ITP 2010. Doi: 10.1007/978-3-642-14052-5_11
- Steen, A. & Benz Müller, C. (2021). **Extensional Higher-Order Paramodulation in Leo-III**. J. Autom. Reasoning 65(6):775–807. Doi: 10.1007/s10817-021-09588-x

